

RESEARCH MONOGRAPH

A Comprehensive Approach to Tail-Risk Estimation

via Extreme Value Theory

*Forecasting and stress-testing warehouse fulfilment losses
in a risk-management setting: a complete mathematical treatment*

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Independent research.

Empirical substrate: UCI Online Retail (Chen, 2015), 1 December 2010 – 9 December 2011.

Companion code: github.com/Shridathj/Warehouse-Anomaly-Detection-Tail-Risk-Stress-Testing

Scope, in one paragraph. This paper derives, from first principles, every estimator used in the warehouse dragon-order tail-risk pipeline: extreme-value tail indices on order quantity, a peaks-over-threshold generalised Pareto model of daily realised loss, log-normal delay simulation, Monte Carlo value-at-risk and expected shortfall, quantile treatment effects, Hawkes maximum likelihood, a Kalman local-linear-trend state-space forecast, and purged expanding-window backtests with Kupiec and Christoffersen tests. Delays are synthetic. Dragon labels are assigned, not observed. Average treatment effects of the surge indicator are therefore a negative control (they are expected to be near zero). The object of causal interest is the quantile contrast in order value. All sterling figures are directional Monte Carlo outputs under stated assumptions; they are not operational loss forecasts.

Abstract

High-value “dragon” orders are rare, heavy-tailed, and operationally expensive when fulfilment service degrades. This paper develops a single mathematical chain from the Fisher–Tippett–Gnedenko theorem to a purged backtest of warehouse tail loss, and instantiates every step on the UCI Online Retail data set under two exposure conventions: gross (maximum booked demand) and customer–SKU netted (demand after cancellations). The ticket file is the UCI Online Retail extract (1 December 2010 – 9 December 2011). It is treated as a real retail order book. It has no pick-complete time, no SLA clock, and no dragon label. Fulfilment delays are therefore simulated from log-normals whose means, dispersion and clips are calibrated to public summaries of the 2025 WERC DC Measures and CSCMP State of Logistics reports — an average-retail delay regime, not a live WMS extract. The diagnostic core is extreme value theory. Order quantities on high-volume SKUs lie in the Fréchet domain: Hill, moment, and GEV block-maxima estimators all return a positive tail index ($\xi \approx 0.47$ – 0.55 gross; 0.48 – 0.83 netted) and the mean-excess plot is linearly increasing. That chapter is a domain diagnosis; the only generalised Pareto MLE in the pipeline is fitted later, to daily realised loss. An AMSE curve is computed as a threshold diagnostic and is not used to fit a GPD on quantity. Annual preventable loss is then a Monte Carlo functional of dragon revenue, an empirical breach rate, and a 30% gross margin. Because dragon assignment is a function of order value, the average treatment effect of a surge flag on net revenue is a negative control and is empirically near zero. The informative contrast is the quantile treatment effect (QTE) of the dragon label on order value. Clustering is modelled by an exponential Hawkes process whose likelihood uses an undecayed-jump recursion; the twelve-month count is projected by a Kalman local-linear-trend filter. A purged expanding-window backtest, with window-matched VaR, is reported together with Kupiec and Christoffersen tests. The paper’s empirical contribution is a reconciliation identity: the several headline sterling numbers in the repository (Monte Carlo ES_{95} , last-window backtest ES_{95} , GPD daily ES, causal annual impact, Hawkes/Kalman P&L bleed) answer different mathematical questions and are not in contradiction once each functional is written down. The scientific limitation is structural: delays are assigned, GPD and Hawkes are fitted in-sample, and one year of public retail tickets is not warehouse telemetry.

Keywords. Extreme value theory; generalised Pareto distribution; Hill estimator; expected shortfall; Hawkes process; Kalman filter; quantile treatment effect; purged backtest; Kupiec test; warehouse fulfilment risk.

Contents

1. Introduction and problem statement
2. Why these methods, and in this order
3. Data, cleaning, and the two exposure conventions
4. Global structure of demand: moments, Pareto, dependence, Gaussianity
5. Extreme value theory, from first principles to the fitted tail
6. Synthetic delays, dragon assignment, holding cost, and SLA loss
7. Monte Carlo value-at-risk and expected shortfall
8. Causal contrasts: ATE as negative control, QTE as the object of interest
9. Hawkes self-exciting point processes
10. State-space projection of monthly dragon counts (Kalman local linear trend)
11. Peaks-over-threshold GPD on daily realised loss
12. Purged expanding-window backtesting
13. Reconciliation of every reported sterling figure
14. Data governance and WMS implications
15. Limitations, and what a production version would change
16. Conclusion
- A. Notation, parameters, and additional derivations
- References

1. Introduction and problem statement

This paper is independent research on tail-risk estimation for warehouse fulfilment. It is written as a technical paper a practising data scientist can read end to end: every estimator is derived, and every sterling headline is identified as a named functional of a named random variable. The empirical substrate is a public retail file, not warehouse telemetry. The conclusions are about the mathematics of a well-specified pipeline, not about a live distribution centre.

1.1 The operational question

A warehouse does not fail on the mean order. It fails on the right tail: a small number of tickets that are simultaneously large in value and slow to fulfil. In the language of this project those tickets are *dragons*—not a zoological claim, but a name for a rare, value-biased sample (about 3.2 basis points of tickets in the gross scenario, 2.5 after netting) that the pipeline treats as the expensive fulfilment tail. Those tickets dominate holding cost, service-level breaches, and the preventable piece of gross-margin loss. The managerial question is narrow and quantitative. If service on those tickets degrades from a 99th-percentile service level to a 95th-percentile service level, how large is the preventable loss, in sterling, over a one-year horizon, and which interventions move that number?

The question is a tail-risk question, not a forecasting-the-mean question. Sample means, ordinary least squares, and Gaussian value-at-risk are the wrong instruments: the quantity distribution on this file has Pearson kurtosis on the order of 10^5 , a coefficient of variation above 13 in the gross scenario, and a maximum ticket of £168,469.60 against a median of £11.80. Extreme value theory exists precisely for this regime. The rest of the pipeline exists to translate a tail index into a loss functional that a risk manager can read: a Monte Carlo distribution of annual preventable loss, a quantile contrast that says how much more valuable dragon tickets are, a point-process model of how dragons cluster, a state-space projection of next year's count, and a backtest that asks whether a VaR number is calibrated.

1.2 What is observed, and what is assigned

The only public file with customer, SKU, quantity, price and timestamp at this scale is the UCI Online Retail data set (Chen, 2015): a UK giftware e-tailer, 1 December 2010 to 9 December 2011. It contains no warehouse timestamps, no pick-path, no SLA clock, and no dragon label. Every delay in this project is therefore a random variable drawn by the analyst, and every dragon label is a weighted sample from the tickets themselves. That fact is not a footnote; it is the identification structure of the paper. It has two immediate consequences.

First, the average treatment effect of the surge indicator on net revenue is a negative control. Surge is assigned independently of covariates once dragons have been removed, and delay given tier is an independent log-normal. There is no causal path from the surge flag to order value, and holding cost is on the order of 100 basis points of value. The ATE must be near zero; empirically it is £7 (gross) and -£21 (netted) on the recorded run. Reporting that number as a policy effect would be a mistake. Second, the quantile treatment effect of the dragon label on order value is partly mechanical: dragons are sampled with probability proportional to a power of value. The QTE is nonetheless the right descriptive object for a risk manager—it answers “how much more valuable is a ticket in the dragon tier, at quantile q ?”—provided it is not sold as an estimated effect of a treatment that nature applied.

1.3 Two scenarios, on purpose

Scenario 1 keeps every positive-quantity ticket after dropping returns. It is a *gross exposure* view: the warehouse must still provision inventory, labour and working capital against booked demand, including tickets that will later be cancelled. Scenario 2 nets quantity at the (CustomerID, SKU) grain over the whole sample and retains only keys with positive net quantity. It is a *fulfilled-demand* view. The famous 80,995-unit “PAPER CRAFT, LITTLE BIRDIE” ticket (StockCode 23843, £168,469.60) is followed within minutes by a matching negative quantity; it dominates Scenario 1 and disappears from Scenario 2. Both views are economically real. Gross exposure is what a WMS sees at order-entry; net demand is what a P&L eventually recognises. A data governance chapter returns to this point.

1.4 Contributions

- A complete derivation of every estimator in the pipeline, from the regular-variation representation of a Fréchet tail through Hill, Dekkers–Einmahl–de Haan, GEV, Pickands–Balkema–de Haan, GPD MLE, AMSE threshold choice, log-normal method of moments, VaR and expected shortfall, Rosenbaum–Rubin propensity scores, Koenker–Bassett quantile regression, Doksum/QTE contrasts, Ogata’s Hawkes likelihood with the implemented undecayed-jump recursion, the Kalman filter for a local linear trend, and the Kupiec and Christoffersen coverage tests.
- An explicit identification statement for the causal chapter: ATE of surge is a negative control; QTE of dragon on value is the object of interest and is partly by construction.
- A reconciliation identity showing that the several sterling headlines in the repository are different functionals of the same simulated file, not competing estimates of one number.

1.5 Organisation

Section 2 motivates the method stack. Section 3 writes down the two data maps. Section 4 records the global diagnostics that rule out Gaussian risk. Section 5 is the EVT core. Sections 6–7 turn the tail into a simulated loss and a Monte Carlo ES. Section 8 is the causal chapter, QTE-first. Sections 9–10 forecast counts. Sections 11–12 backtest. Section 13 reconciles money. Sections 14–16 discuss data governance, limits, and what the work does and does not show.

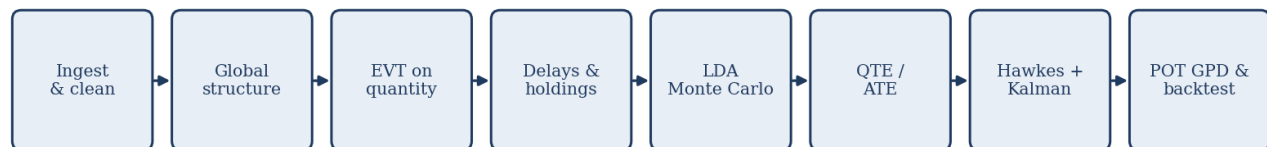


Figure 1. The pipeline as a directed chain. Later sterling numbers are functionals of earlier objects; they should be read as a family, not as a single “the” loss.

2. Why these methods, and in this order

2.1 Why not a Gaussian, and why not a black-box detector

Anomaly detection in operations research often means an isolation forest, an autoencoder, or a z-score on a rolling mean. Those tools answer a different question: “which tickets look unlike the others?” The present question is “how heavy is the right tail, and what is the loss functional of that tail under a service-level

shock?” A z-score assumes a second moment that is a useful scale. Here the fourth moment is effectively infinite for practical sample sizes (Pearson kurtosis 178,189 on gross quantity). Isolation forests do not produce a tail index, a mean-excess function, or an expected shortfall. Extreme value theory does.

2.2 Extreme value theory, first

EVT is used first because it is the only asymptotically justified model for the far right tail of i.i.d. (or weakly dependent) observations. The Fisher–Tippett–Gnedenko theorem classifies the limit law of maxima. The Pickands–Balkema–de Haan theorem classifies the limit law of excesses over a high threshold. Together they give a one-parameter diagnosis—the sign of ξ —that decides whether the warehouse is in a heavy-tailed world (Fréchet), an exponential world (Gumbel), or a bounded world (Weibull). The quantity chapter uses that diagnosis only: Hill, the moment estimator, GEV block maxima, and a mean-excess slope. A GPD is not fitted to quantity. The only GPD MLE in the repository is the daily-loss fit of Section 11. If $\xi \leq 0$ on quantity, dragons are not a regularly varying phenomenon and the rest of the stack should be rebuilt.

2.3 Simulation, because the SLA clock is missing

Given a tail, one still needs a delay. The file has InvoiceDate, not pick-complete time. The honest move is to simulate delays from a parametric family whose means, coefficients of variation and clip ranges are taken from public WERC DC Measures and CSCMP State of Logistics summaries, and to say so. A log-normal is used because cycle times are positive, right-skewed, and conventionally modelled that way in fulfilment analytics. Dragons are not a random 3.2 bps sample: they are value-biased, because the economic object of interest is the expensive tail, not a uniform random ticket that happens to be slow. The 99th→95th language is SLA / QoS degradation on that expensive tail: configured dragon delays sit in a degraded service band relative to a tight on-time SLA. It is not a causal estimate of a live distribution centre.

2.4 Monte Carlo, because the annual loss is a sum of rare events

Annual preventable loss is a sum, over 365 days, of a random daily dragon revenue times an empirical breach rate times a margin. That sum has no useful closed form once daily revenue is log-normal and the breach rate is a ratio of two sample totals. Monte Carlo is the standard way to read VaR and expected shortfall off a distribution one can sample. The i.i.d. assumption is a computational convenience; clustering is handled in a separate Hawkes chapter and is not folded into the Monte Carlo paths. That separation is deliberate (the Loss Distribution Approach tradition of a marginal frequency–severity product, with a Hawkes-modulated year left as future work).

2.5 QTE rather than ATE

A mean contrast between surge and non-surge tickets, after propensity matching, estimates the wrong thing: surge is assigned, not discovered, and it does not cause order value. A quantile contrast between dragon and non-dragon order values estimates the object a risk manager actually needs—how the value distribution of the flagged tail sits relative to the rest—at the price of admitting that the flag is itself a function of value. Section 8 writes both estimators down and uses the ATE only as a specification check.

2.6 Hawkes, then a Kalman projection

Dragons are events in time. If they cluster, a Poisson baseline understates short-horizon intensity. A Hawkes process with exponential kernel is the simplest self-exciting model with a closed compensator and a recursive likelihood. The implementation maximises that likelihood under an undecayed-jump recursion, described in Section 9. A twelve-month count forecast, by contrast, is a small time-series problem (thirteen monthly points). A local linear trend Kalman filter is the linear-Gaussian tool for that job. It is not a Bayesian

structural time series: there is no posterior over Q and R , no spike-and-slab, no PyMC.

2.7 Purged backtests, because leakage is the default

Fitting a tail model on the whole sample and then scoring the whole sample is not a test. An expanding window with a purge gap between train cutoff and test end is the standard way, in quantitative risk, to respect serial dependence and overlapping information (cf. López de Prado's purged cross-validation). Kupiec's unconditional coverage test and Christoffersen's independence test are the textbook pair for VaR exception sequences. With one year of data their power is low; that is stated, not hidden.

3. Data, cleaning, and the two exposure conventions

3.1 The file

Let a raw row be a tuple (InvoiceNo, StockCode, Quantity, InvoiceDate, UnitPrice, CustomerID, Country). The UCI Online Retail file contains 541,909 such rows. CustomerID is missing for a non-trivial share of tickets; those rows are dropped because Scenario 2's netting key requires an identified customer. Unit prices that are non-positive are dropped. After this filter the working sample for Scenario 1 is 397,884 positive-quantity tickets, 3,665 SKUs, spanning 374 calendar days, with gross merchandise value £8,911,407.90.

3.2 Scenario 1: gross booked demand

Write Q_i for quantity and P_i for unit price. Scenario 1 keeps rows with $Q_i > 0$, $P_i > 0$ and CustomerID observed, and defines order value by

$$V_i = Q_i P_i. \quad (1)$$

No cancellation is netted. A subsequent negative-quantity row with the same customer and SKU is simply absent from this file. Gross value is therefore an upper bound on recognised demand and a realistic bound on short-horizon operational exposure: pickers, inventory and working capital are committed when the ticket is booked, not when it survives the returns window.

3.3 Scenario 2: customer-SKU netting

Let the netting key be $\kappa = (\text{CustomerID}, \text{SKU})$, concatenated in the code as a string. Let G be the set of gross picks ($Q > 0$) and C the set of cancellations ($Q < 0$), both after the UnitPrice and CustomerID filter. Define

$$N_\kappa = \sum_{i: \kappa(i) = \kappa} Q_i. \quad (2)$$

Retain keys with $N_\kappa > 0$. For each such key the code keeps a single representative row: the gross pick of maximum quantity on that key, and overwrites its quantity and value by the net,

$$Q_{i^*(\kappa)} \leftarrow N_\kappa, \quad V_{i^*(\kappa)} \leftarrow N_\kappa P_{i^*(\kappa)}, \quad (3)$$

where $i^*(\kappa) = \arg \max \{Q_i : i \in G, \kappa(i)=\kappa\}$. The resulting table has 265,210 rows (8,905 cancellation rows participated in the netting). This is a strong identifying assumption: all cancellations of a customer-SKU pair, at any time in the year, are charged against that customer's largest pick of that SKU, and the timestamp of the kept row is the timestamp of that largest pick. It is not invoice-level netting, and it is not a twelve-minute matching window. It is an annual, key-level, max-pick representative.

3.4 Why both conventions must exist

If the 80,995-unit ticket is a system glitch, Scenario 1 double-counts a phantom and Scenario 2 is the truth. If it is a real order that was cancelled after the warehouse had already allocated stock, Scenario 1 is the exposure that risk management is supposed to see, and Scenario 2 is the accounting residual. A production WMS would resolve the ambiguity with an order-entry audit trail. On this public file one cannot. Reporting both scenarios is the substitute for that audit trail.

Quantity	Scenario 1 (gross)	Scenario 2 (netted)
Rows after cleaning	397,884	265,210
Unique SKUs	3,665	3,649
Gross / net merchandise value	£8,911,407.90	£8,307,997.38
Mean ticket value	£22.40	£31.33
Median ticket value	£11.80	£14.75
99th percentile value	£202.50	£316.15
99.9th percentile value	£876.00	£1,756.30
Maximum ticket	£168,469.60	£16,588.00
Mean quantity (unfiltered)	12.99	18.53
CV of quantity	13.807	5.010
Pearson kurtosis of quantity	178,189	3,324

Table 1. The two working samples. Netting removes the £168k PAPER CRAFT ticket and collapses a large mass of cancellations; the remaining net tickets are fewer, larger on average, and still heavy-tailed.

4. Global structure of demand: moments, Pareto, dependence, Gaussianity

4.1 Location, scale, shape

Let $Q_1, \dots, Q_n$ be transaction quantities. The pipeline reports both the unfiltered sample and a Pareto-filtered subsample. Write

$$CV = s/\bar{Q}, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (Q_i - \bar{Q})^2. \quad (4)$$

The coefficient of variation is dimensionless and is the natural scale-free dispersion for a positive variable. A CV of 13.8 (gross, unfiltered) means the standard deviation is an order of magnitude larger than the mean; the mean is not a typical ticket. Skewness and Pearson kurtosis are

$$\gamma_1 = \frac{m_3}{s^3}, \quad \kappa = \frac{m_4}{s^4}, \quad m_r = \frac{1}{n} \sum_i (Q_i - \bar{Q})^r. \quad (5)$$

`pandas.Series.kurtosis` returns excess kurtosis (Fisher); the code adds 3 to report Pearson kurtosis κ , which equals 3 for a Gaussian. On the unfiltered gross file $\kappa = 178,189$. That single number is the argument against every Gaussian risk measure that will be written down later.

4.2 The 80/20 filter

Let S be the set of SKUs and let $T_s = \sum_{i: \text{SKU}(i)=s} Q_i$ be total quantity of SKU s . Rank SKUs so that $T_{(1)} \geq T_{(2)} \geq \dots$ and form the cumulative share

$$C_k = \frac{\sum_{j=1}^k T_{(j)}}{\sum_{s \in S} T_s}. \quad (6)$$

The high-volume set is $\{s : C_{k(s)} \leq 0.8\}$, i.e. the largest prefix of the ranked list whose cumulative share is at most 80% of total quantity (the code's `cum_freq <= 0.8`; it can undershoot 80% by one SKU). Tail estimators are then computed on the transaction-level quantities of those SKUs, not on the SKU totals. The filter is a concentration device: it asks whether the tail of operationally important SKUs is heavy, not whether a long tail of one-off novelty SKUs is heavy.

Scenario 1 ranks SKUs by sum of quantity. Scenario 2's EVT chapter ranks SKUs by maximum single-order quantity and retains those accounting for 80% of the sum of those maxima (797 SKUs, $n = 131,137$ tickets). Scenario 2 is looking at SKUs capable of a large single net order, which is the relevant grain after netting. Table 2's Scenario 2 diagnostics are computed under the sum filter in `global_stats.py`; Table 3's Scenario 2 tail indices use the max filter in `run_evt_gpd`.

4.3 Serial dependence of daily totals

Let Y_t be total quantity on calendar day t , $t = 1, \dots, T$. The sample autocorrelation at lag k is

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^T (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^T (Y_t - \bar{Y})^2}. \quad (7)$$

Ljung–Box (1978) tests the joint hypothesis $\rho_1 = \dots = \rho_h = 0$ with

$$Q_{\text{LB}} = T(T+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{T-k} \quad (8)$$

under white noise. With $h = 10$ the pipeline reports $Q_{\text{LB}} = 136.48$ ($p \approx 0$) on gross daily totals and $Q_{\text{LB}} = 40.76$ ($p \approx 0$) after netting. Daily volume is not white. EVT theorems are typically stated for i.i.d. observations or for stationary sequences with mixing; the practical consequence is that tail-index standard errors computed under i.i.d. are optimistic, and that a Hawkes chapter is not an ornament. Serial dependence is left in the series on purpose (no declustering): Hill / GEV / POT remain i.i.d. estimators, used here to show that the data are both dependent and extreme.

4.4 Three Gaussianity tests, and why they all reject

Kolmogorov–Smirnov. Standardise $Z_i = (Q_i - Q)/s$ and let F_n be the empirical distribution function of the Z_i . The KS statistic against the standard normal cdf Φ is

$$D_n = \sup_x |F_n(x) - \Phi(x)|. \quad (9)$$

The code calls `scipy.stats.kstest` on the z-scores. Because mean and variance are estimated, the tabulated KS p -value is the Lilliefors situation: the test is liberal. It does not matter here. $D_n = 0.473$ (gross) and 0.421 (netted); even a correctly sized test would reject at any conventional level.

Anderson–Darling. The AD statistic weights the tails of the empirical process,

$$A^2 = n \int_{-\infty}^{\infty} \frac{(F_n(x) - F_0(x))^2}{F_0(x)(1 - F_0(x))} dF_0(x). \quad (10)$$

`statsmodels.stats.normal_ad` returns $A^2 = \infty$ on this file. With outliers on the order of 80,000 against a mean of 13, $F_0(x)(1 - F_0(x))$ in the far tail is smaller than machine precision relative to the squared discrepancy, and the integral overflows. It is the correct qualitative message: the Gaussian tail is not a model for these data.

Shapiro–Wilk. The SW statistic is the squared correlation between ordered observations and normal scores, equivalently

$$W = \frac{\left(\sum_{i=1}^n a_i Q_{(i)} \right)^2}{\sum_{i=1}^n (Q_i - \bar{Q})^2}, \quad (11)$$

with a constructed from the mean and covariance of normal order statistics. W is computed on a without-replacement subsample of 5,000 drawn from `default_rng(314159)` (the Royston approximation is not reliable at $n \approx 4 \times 10^5$). On the recorded run $W = 0.202$ (gross) and 0.055 (netted); a Gaussian has $W \approx 1$.

4.5 A first look at the tail, before EVT

The empirical mean excess at the 99th percentile of the Pareto-filtered quantities is already a heavy-tail smoking gun. With $u = Q_{0.99}$,

$$e_n(u) = \frac{1}{N_u} \sum_{i: Q_i > u} (Q_i - u). \quad (12)$$

One obtains $e_n(144) = 288.1$ (gross) and $e_n(276) = 469.8$ (netted). For an exponential random variable the mean excess is constant and equal to the scale; a mean excess twice the threshold is the signature of a regularly varying tail, which Section 5 formalises.

Diagnostic	Gross (S1)	Netted (S2)	Gaussian benchmark
Mean quantity (unfiltered)	12.99	18.53	—
CV	13.807	5.010	s/μ
Skewness	409.89	43.69	0
Pearson kurtosis	178,189	3,324	3
ACF lags 1–5 (daily totals)	0.27, 0.15, 0.16, 0.16, 0.20	0.18, 0.15, 0.10, 0.14, 0.12	0
Ljung–Box Q (lag 10)	136.48 (p=0)	40.76 (p=0)	χ^2_{10}
KS D (z-scores)	0.473 (p=0)	0.421 (p=0)	0
Shapiro–Wilk W (n=5,000)	0.202 (p=0)	0.055 (p=0)	1
99th pctl quantity	144	276	—
Mean excess at 99th pctl	288.1	469.8	scale

Table 2. Global diagnostics. Every Gaussian and every white-noise benchmark is rejected. The tail chapter is not optional. KS, Shapiro–Wilk, the 99th percentile and the mean excess use the Pareto-filtered quantity vector; mean, CV, skewness and kurtosis in the first four rows are unfiltered.

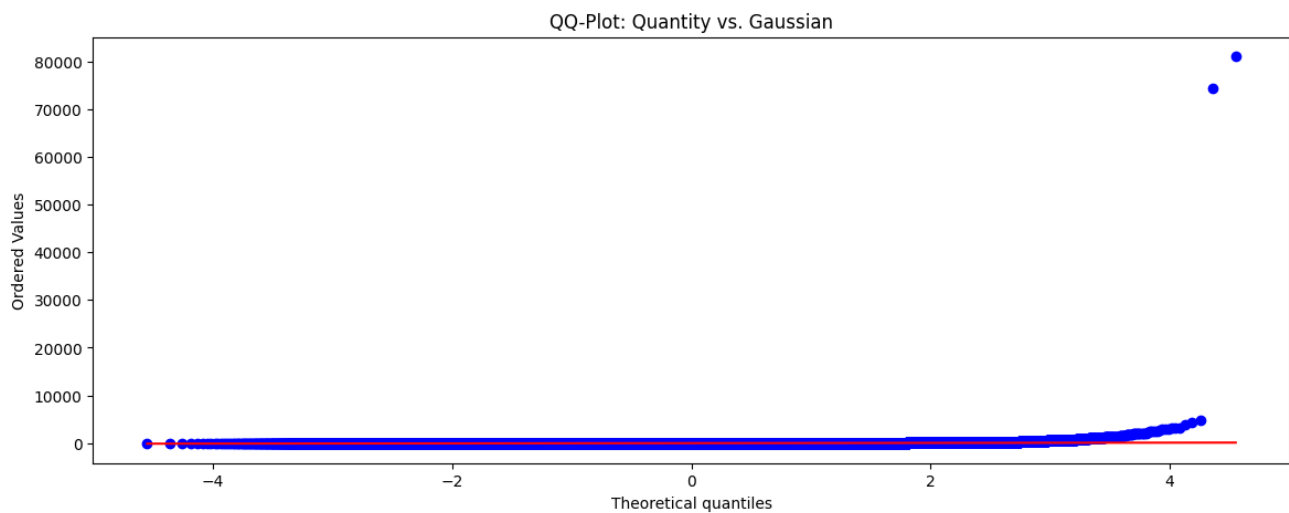


Figure 2. Normal Q-Q-plot of Pareto-filtered quantities, Scenario 1. The right-tail bend is the Fréchet signature in a Gaussian plot.

5. Extreme value theory, from first principles to the fitted tail

5.1 What EVT is for

Let $X_1, \dots, X_n$ be i.i.d. with distribution function F , and write $M_n = \max\{X_1, \dots, X_n\}$. For any F with a finite right endpoint or a thin tail, M_n is not interesting to a risk manager: sample maxima concentrate. For a heavy-tailed F , M_n grows and the way it grows is classified by a single real parameter ξ , the extreme-value index (tail index). EVT is the limit theory of M_n and, equivalently, of excesses $X - u$ given $X > u$ as u tends to the right endpoint. This section derives the three limit laws, the Hill estimator as the MLE of an exact Pareto tail, the Dekkers–Einmahl–de Haan moment estimator, GEV fitting of block maxima, the Pickands–Balkema–de Haan theorem, the GPD likelihood (used later on daily loss, not on quantity), the mean-excess diagnostic, an AMSE threshold diagnostic, and the empirical numbers on this file.

5.2 Regular variation

Definition (Regular variation at infinity).

A survival function $1 - F$ is regularly varying with index $-1/\xi$, $\xi > 0$, if there exists a slowly varying function L such that $1 - F(x) = x^{-1/\xi} L(x)$ for x large. Slowly varying means $L(tx)/L(x) \rightarrow 1$ as $x \rightarrow \infty$ for every $t > 0$. Equivalently, $(1 - F(tx))/(1 - F(x)) \rightarrow t^{-1/\xi}$. The number $\alpha = 1/\xi$ is the tail exponent: $P(X > x)$ behaves like $x^{-\alpha}$.

If $\xi > 1/2$ then $\text{Var}(X) = \infty$; if $\xi \geq 1$ then $E[X] = \infty$. The Hill estimates on this file sit near $\xi \approx 0.5$, i.e. on the knife-edge of infinite variance. That is why a sample standard deviation of quantity is a fragile object, and why expected shortfall, which is a tail mean, must be handled through a GPD rather than through a sample mean of the whole distribution.

5.3 Fisher–Tippett–Gnedenko

Suppose there exist affine normalisations $a_n > 0$, $b_n \in \mathbb{R}$ such that

$$P((M_n - b_n)/a_n \leq z) \rightarrow G(z) \quad (13)$$

for a non-degenerate G . Then G belongs to one of three families, which can be written as a single generalised extreme value (GEV) law (von Mises–Jenkinson):

$$G_\xi(z) = \exp\left(- (1 + \xi z)_+^{-1/\xi}\right), \quad (14)$$

with the $\xi \rightarrow 0$ case defined by continuity as $G_0(z) = \exp(-e^{-z})$. The sign of ξ is the whole theory:

- $\xi > 0$, Fréchet. Heavy tails, regularly varying survival, unbounded support. Maxima grow like n^ξ . This is the empirically relevant case here.
- $\xi = 0$, Gumbel. Light tails of exponential type (Gaussian, log-normal, exponential, gamma). Mean excess of the parent is asymptotically constant or slowly varying.
- $\xi < 0$, Weibull. Finite right endpoint. Excesses shrink linearly toward that endpoint. A negative ξ on daily loss (Section 11, Scenario 2) is possible even when quantity is Fréchet, because daily loss is a sparse compound of a few unfulfilled dragons.

Theorem (Fisher–Tippett–Gnedenko, 1928/1943/1948).

The only possible non-degenerate limit laws of affinely normalised i.i.d. maxima are the GEV family $\{G_\xi : \xi \in \mathbb{R}\}$. F is said to lie in the max-domain of attraction of G_ξ , written $F \in \text{MDA}(G_\xi)$.

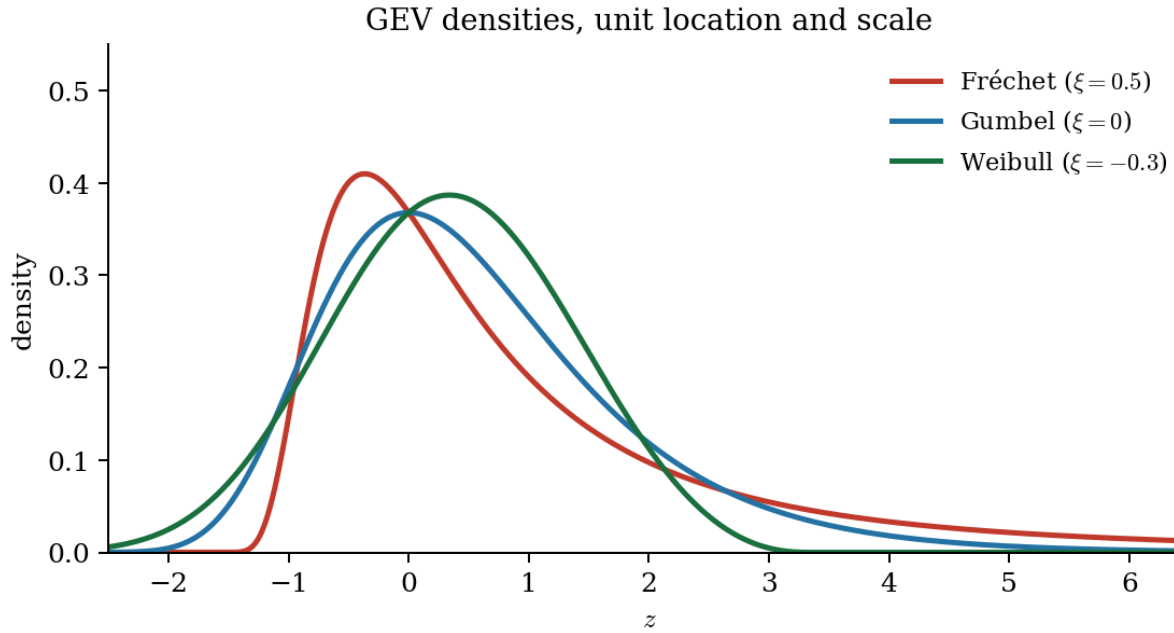


Figure 3. GEV densities in the three domains, unit location and scale. Fréchet (red) has a polynomial right tail; Gumbel (blue) is exponential; Weibull (green) is bounded on the right.

5.4 Why scipy's c is $-\xi$

scipy.stats.genextreme uses the convention of the National Institute of Standards, in which the shape is $c = -\xi$ and the cdf is $\exp(-(1 - cz)^{1/c})$ for $c \neq 0$. The pipeline therefore reports

$$\hat{\xi}_{\text{GEV}} = -\hat{c}. \quad (15)$$

A positive printed ξ_{GEV} is a negative scipy c , i.e. Fréchet. This is the Coles (2001) sign convention, and it is the one used throughout this paper.

5.5 Hill's estimator, derived as a Pareto MLE

Condition on $X > u$ and suppose the parent, not the excess $Y = X - u$, is exactly Pareto with tail index ξ : $P(X > x \mid X > u) = (x/u)^{-1/\xi}$ for $x > u$. (The excess of a GPD is a different object, used later on daily loss.) The log-likelihood of the k observations $x_1, \dots, x_k$ above u is

$$\ell(\xi) = -k \log \xi - \frac{1}{\xi} \sum_{j=1}^k \log(x_j/u) - k \log u. \quad (16)$$

The $-k \log u$ term does not depend on ξ . Differentiate the rest and set the score to zero to obtain the closed form

$$\hat{\xi}_{\text{Hill}} = \frac{1}{k} \sum_{i=1}^k \log \frac{X_{(n-i+1)}}{X_{(n-k)}}, \quad (17)$$

where $X_{(1)} \leq \dots \leq X_{(n)}$ are the order statistics and $u = X_{(n-k)}$ is the $(k+1)$ -st upper order statistic. This is Hill (1975). Under regular variation with index $-1/\xi$ and $k \rightarrow \infty$, $k/n \rightarrow 0$, one has $\hat{\xi}_{\text{Hill}} \rightarrow_p \xi$, and $\sqrt{k}(\hat{\xi}_{\text{Hill}} - \xi)$ is asymptotically normal with variance ξ^2 in the exact Pareto case.

The pipeline takes $k = \lfloor \sqrt{n} \rfloor$ with n the filtered transaction count (in Scenario 1 the EVT call uses the high-volume quantity vector, so $k = \lfloor \sqrt{n} \rfloor$ on that length), sorts the Pareto-filtered quantities in decreasing order, and evaluates

$$\hat{\xi}_{\text{Hill}} = \frac{1}{k} \sum_{i=1}^k \log (X_{[i]} / X_{[k+1]}), \quad (18)$$

where $X_{[i]}$ is the i -th decreasing value. Empirically $\hat{\xi}_{\text{Hill}} = 0.515$ (gross) and 0.602 (netted). Both are strictly positive: Fréchet. The implied tail exponents are $\alpha = 1/\hat{\xi} \approx 1.94$ (gross) and 1.66 (netted). In both cases $\alpha < 2$, so a finite-variance model of quantity is not justified asymptotically.

5.6 The moment estimator of Dekkers, Einmahl and de Haan

Hill is designed for $\xi > 0$. The moment estimator (Dekkers, Einmahl and de Haan, 1989) is consistent for $\xi \in \mathbb{R}$. Write $M_n^{(1)}$ for the Hill sum and

$$M_n^{(2)} = \frac{1}{k} \sum_{i=1}^k \left(\log \frac{X_{[i]}}{X_{[k+1]}} \right)^2. \quad (19)$$

For $\xi > 0$ the log-excesses are asymptotically exponential with mean ξ , so $M_n^{(1)} \approx \xi$ and $M_n^{(2)} \approx 2\xi^2$: the ratio $(M_n^{(1)})^2 / M_n^{(2)}$ converges in probability to $1/2$, the second-moment correction vanishes, and the estimator is asymptotically Hill. For general ξ the estimator

$$\hat{\xi}_{\text{Mom}} = M_n^{(1)} + 1 - \frac{1}{2} \left(1 - (M_n^{(1)})^2 / M_n^{(2)} \right)^{-1} \quad (20)$$

is the form implemented in the code (the factor $\frac{1}{2}(1 - \text{ratio})^{-1}$ is the standard second-moment correction; the code requires $\text{ratio} < 1$). On this file $\hat{\xi}_{\text{Mom}} = 0.551$ (gross) and 0.483 (netted). Agreement of Hill and moment, both positive, is the first robustness check that the Fréchet diagnosis is not an artefact of Hill's Pareto assumption.

5.7 Block maxima and a GEV fit

Partition the sample into m blocks of length b , $b \approx n$, and let $M_j^{(b)}$ be the maximum in block j . If $F \in \text{MDA}(G_\xi)$ then the block maxima are approximately GEV. The log-likelihood of an i.i.d. GEV sample is

$$\ell(\xi, \mu, \sigma) = -m \log \sigma - (1 + 1/\xi) \sum_j \log(1 + \xi z_j) - \sum_j (1 + \xi z_j)^{-1/\xi} \quad (21)$$

for $\xi \neq 0$, with $z_j = (M_j^{(b)} - \mu)/\sigma$ and the Gumbel limit obtained by L'Hôpital. Maximisation is numerical (scipy.stats.genextreme.fit uses a combination of method of moments and MLE).

The code sets `block_size = 30` and walks through the quantity array of high-volume SKUs in file order, taking the maximum of each complete block of 30 consecutive filtered transactions (blocks of 30 consecutive tickets, not calendar months). The fitted $\xi_{\text{GEV}} = 0.473$ (gross) and 0.833 (netted). The netted GEV is the most dispersed of the three estimators, which is expected: block maxima of size 30 on a heavy tail are noisy, and Scenario 2's filter on maxima inflates block extremes.

5.8 Peaks over threshold: Pickands–Balkema–de Haan

Block maxima throw away all but one observation per block. The peaks-over-threshold (POT) method keeps every excess over a high threshold u . Let $x_F = \sup\{x : F(x) < 1\}$ be the right endpoint (possibly $+\infty$).

Theorem (Pickands–Balkema–de Haan).

$F \in \text{MDA}(G_\xi)$ if and only if there exists a positive measurable function $\sigma(u)$ such that the conditional excess distribution $F_u(y) := P(X - u \leq y \mid X > u)$ satisfies $\sup_{y \in (0, x_F - u)} |F_u(y) - G_\xi\{\xi, \sigma(u)\}(y)| \rightarrow 0$ as $u \rightarrow x_F$, where $G_\xi\{\xi, \sigma\}$ is the generalised Pareto distribution (GPD).

The GPD cdf, density and survival, for $y \geq 0$ and $1 + \xi y/\beta > 0$, are

$$G(y) = 1 - (1 + \xi y/\beta)_+^{-1/\xi}, \quad g(y) = \beta^{-1} (1 + \xi y/\beta)_+^{-1/\xi - 1}. \quad (22)$$

For $\xi \rightarrow 0$ this is the exponential law with mean β . For $\xi > 0$ it is Pareto type II (Lomax) with unbounded support. For $\xi < 0$ it has a finite endpoint $y = -\beta/\xi$. The scale used in the code for daily-loss POT is written β rather than σ ; this paper uses β for the GPD scale to avoid clashing with Hawkes decay.

5.9 Mean excess, and why a positive slope is the GPD's signature

The mean excess function $e(u) = E[X - u \mid X > u]$, when it exists ($\xi < 1$), is, for an exact GPD,

$$e(u) = \frac{\beta + \xi u}{1 - \xi}, \quad \xi < 1. \quad (23)$$

A scatter of empirical mean excesses against u that is linearly increasing in the upper tail is therefore visual evidence for $\xi > 0$. The pipeline computes

$$e_n(u) = \frac{1}{\#\{i: X_i > u\}} \sum_{i: X_i > u} (X_i - u) \quad (24)$$

on a grid of empirical percentiles from the 90th to the 99th (40 points), then fits a least-squares slope on the upper 40% of that grid. The fitted slopes are +1.782 (gross) and +1.089 (netted). Both are positive. If one treats the slope as $\xi/(1-\xi)$, one recovers a rough $\xi \approx 0.64$ (gross) and 0.52 (netted), in the same neighbourhood as Hill. This is a diagnostic, not an estimator: the least-squares slope on a dependent grid is not Hill.

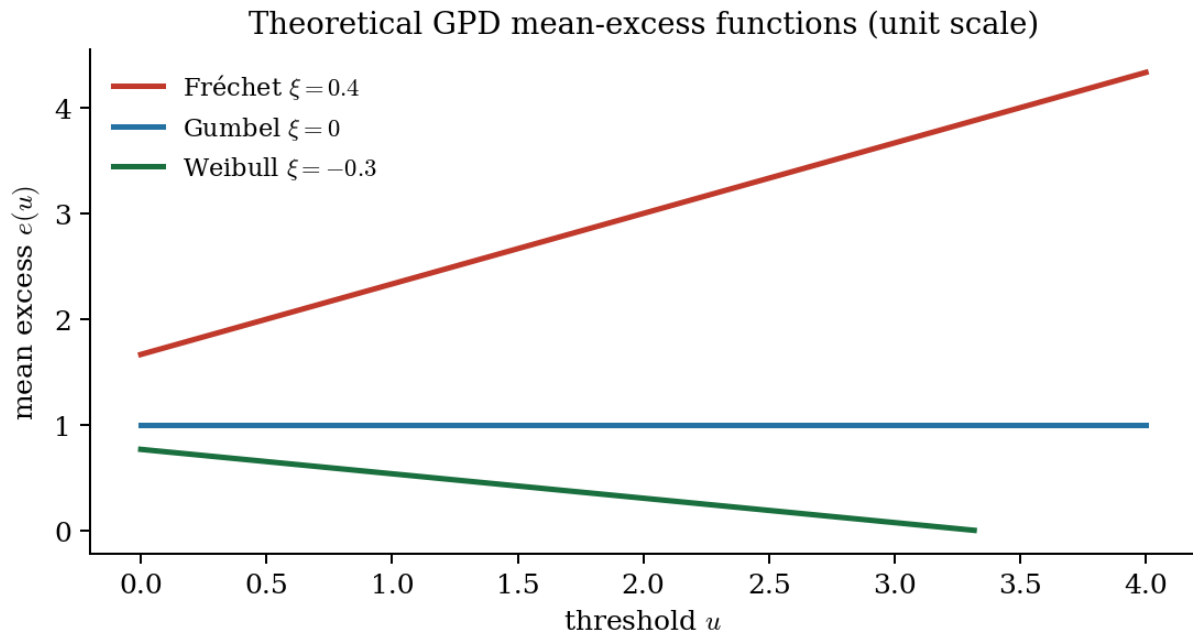


Figure 4. Theoretical mean-excess functions of a GPD with unit scale. Only the Fréchet case ($\xi > 0$) produces a linearly increasing mean excess. That is the pattern in both empirical plots of the repository.



Figure 5. Empirical mean-excess plot, Scenario 1, Pareto-filtered quantity. The upper-tail least-squares slope is +1.782.

5.10 GPD maximum likelihood

Given exceedances $Y_j = X_j - u > 0, j = 1, \dots, N_u$, the GPD log-likelihood is

$$\ell(\xi, \beta) = -N_u \log \beta - (1/\xi + 1) \sum_{j=1}^{N_u} \log(1 + \xi Y_j / \beta), \quad (25)$$

provided $\beta > 0$, $\xi > -1$, and $1 + \xi Y_j / \beta > 0$ for all j . Asymptotic normality of the MLE additionally wants $\xi > -1/2$. Expected shortfall of the parent is finite only for $\xi < 1$. The negative log-likelihood implemented for daily losses in the backtest module is exactly $-\ell$, with a barrier that returns a large constant when $\beta \leq 0$, $\xi \geq 1$, or any term $1 + \xi Y_j / \beta$ is non-positive, and with box bounds $\xi \in [-0.5, 1]$. The code rejects $\xi \geq 1$: ES of a GPD parent is infinite there, so the optimiser is kept out of the non-integrable region. This likelihood is not fitted to quantity. The quantity chapter is a Fréchet-domain diagnosis. The only GPD likelihood in the repository is the daily-loss fit of Section 11.

5.11 AMSE-optimal threshold

Threshold choice is the practical problem of POT. Too low a u contaminates the GPD approximation (bias); too high a u starves the likelihood (variance). For each candidate threshold the code forms the excesses $Y = X - u$, sets $k = N_u$ (every excess), and evaluates a Hill-type mean of $\log(Y_{[i]} / Y_{\min})$ on i.i.d. bootstrap resamples of those excesses, using `default_rng(314159)` and $B = 200$. A bootstrap estimate of asymptotic mean squared error is

$$\widehat{\text{AMSE}}(u) = \widehat{\text{Bias}}_u^2 + \widehat{\text{Var}}_u, \quad (26)$$

where the bias is measured against the full-sample Hill of Section 5.5. The code evaluates this on the percentile grid $\{92, \dots, 98.5\}$ (25 points), skips any u with fewer than 50 excesses, and takes $u^* = \arg \min \widehat{\text{AMSE}}(u)$ over the valid grid. Empirically $u^* = 96.0$ (gross) and $u^* = 49.0$ (netted).

AMSE here is a stability criterion, not an oracle MSE. The bootstrap of excesses is i.i.d. and ignores the daily dependence documented in Section 4.3. Operationally: u^* is never passed to a GPD MLE on quantity. The quantity chapter is a Fréchet-domain diagnosis. None of this overturns the qualitative conclusion that a regularly varying tail, not a Gaussian, is the right family for quantity.

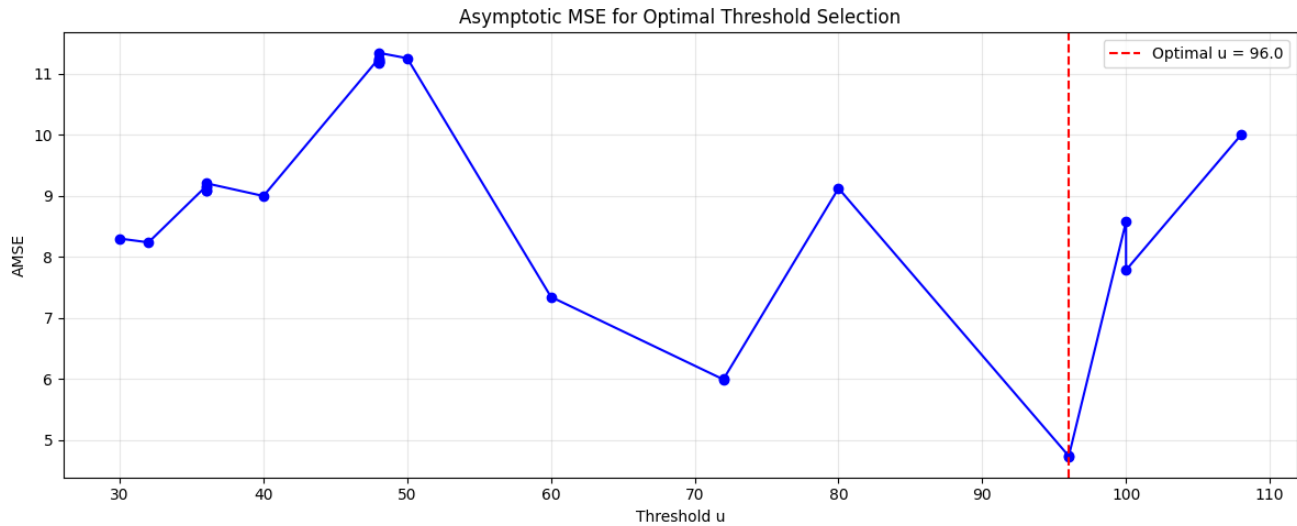


Figure 6. Bootstrap AMSE of the excess-Hill statistic as a function of threshold, Scenario 1. The vertical line is the minimiser $u^* = 96$. That threshold is not used to fit a GPD on quantity.

5.12 Empirical tail-index table, and the Fréchet conclusion

Estimator	S1 gross	S2 netted	What it is
Hill ($k = \lfloor \sqrt{n} \rfloor$)	0.515	0.602	MLE of an exact Pareto tail
Moment	0.551	0.483	Dekkers–Einmahl–de Haan
GEV block-max	0.473	0.833	MLE on 30-transaction blocks
Mean-excess slope	+1.782	+1.089	LS on upper 40% of $e_n(u)$
Implied ξ from slope	≈ 0.64	≈ 0.52	slope = $\xi/(1-\xi)$
AMSE threshold u^*	96.0	49.0	min excess-Hill AMSE
Domain	Fréchet	Fréchet	all $\xi > 0$ on quantity

Table 3. Tail-index estimates on high-volume SKU quantities. Quantity is in the Fréchet domain in both scenarios. Daily-loss GPD in Section 11 is a different random variable and can have a different ξ .

The paper’s EVT conclusion is therefore modest and firm. Transaction quantity, on the SKUs that constitute 80% of volume (S1) or 80% of max-order mass (S2), has a regularly varying right tail with ξ in a neighbourhood of one half. Gaussian, exponential, and bounded-tail models for quantity are not compatible with this evidence. That is the licence for everything that follows: a dragon is not an outlier to be winsorised; it is a point in a regularly varying tail, and the loss it can generate has to be read with tail tools.

Remark. Convergence of three estimators is not a proof of i.i.d. regular variation. It is a check that the Fréchet diagnosis is not an artefact of one functional. Dependence (Section 4.3), the 30-transaction GEV blocking, and the value-biased sampling of dragons later in the pipeline all mean that ξ should be read as a directional tail index, not as a number to three decimals for a regulator.

6. Synthetic delays, dragon assignment, holding cost, and SLA loss

6.1 Why the delays are synthetic

InvoiceDate is a billing timestamp, not a fulfilment clock. There is no pick-start, no pack-complete, no carrier handover, and no SLA flag in the UCI file. Any delay attached to a ticket is therefore a random

variable constructed by the analyst. The construction is parametric and documented in `src/config.py`. Parameters are educated values informed by public summaries of the 2025 WERC DC Measures Report and the CSCMP State of Logistics Report, not extractions from those proprietary files. This section writes the construction down so that the invented randomness is fully specified.

6.2 Value-biased dragon assignment

Let n be the number of tickets and let p_d be the dragon rate (0.00032 gross, 0.00025 netted). Let γ be the bias exponent (1.35 gross, 1.20 netted). Define sampling weights

$$w_i = V_i^\gamma / \sum_{j=1}^n V_j^\gamma. \quad (27)$$

A set D of size $n_d = \lfloor p_d n \rfloor$ is drawn without replacement from $\{1, \dots, n\}$ with probabilities w . (numpy's choice with `p=w` and `replace=False` implements a sequential weighted sample; inclusion probabilities are not exactly w_i , but they are monotone in V_i and approximately w_i for small p_d .) The remaining tickets contribute a surge set S of size $\lfloor 0.18 n \rfloor$ drawn uniformly, and the rest are labelled normal. Write $T_i \in \{\text{normal}, \text{surge}, \text{dragon}\}$ for the resulting tier. Empirically $|D| = 127$ (0.0319%) in Scenario 1 and 66 (0.0249%) in Scenario 2. The generator is `default_rng(314159)`.

Proposition (Stochastic dominance, mechanical).

If $\gamma > 0$ and sampling is value-biased as above, then $V \mid T=\text{dragon}$ likelihood-ratio dominates $V \mid T \neq \text{dragon}$ in the large- n weighted-sampling limit. Consequently $Q_q(V \mid \text{dragon}) - Q_q(V \mid \text{non-dragon}) \geq 0$ at every quantile q , with strict inequality on a set of q of positive measure. The QTE of Section 8 is therefore partly a restatement of the assignment rule. It is still the right descriptive contrast for a risk manager, because the assignment rule was chosen to put the expensive tail in the dragon bucket; it is not an estimated effect of an external treatment.

6.3 Log-normal delays, parameterised by the mean

Given a tier, delay D (minutes) is drawn independently from a log-normal and then clipped to a closed interval. A log-normal is specified by (μ_ℓ, σ_ℓ) , but the configuration file specifies a mean in minutes m and a log-scale σ_ℓ . The translation is the elementary moment identity. If $\log D_0 \sim N(\mu_\ell, \sigma_\ell^2)$ then

$$E[D_0] = \exp(\mu_\ell + \frac{1}{2}\sigma_\ell^2). \quad (28)$$

Imposing $E[D_0] = m$ gives $\mu_\ell = \log m - \frac{1}{2}\sigma_\ell^2$, which is exactly the line in `delays.py`:

$$D = \text{clip}(\exp(\mu_\ell + \sigma_\ell Z), a, b), \quad Z \sim N(0, 1). \quad (29)$$

Clipping truncates the log-normal and therefore lowers the realised mean relative to m , more so for dragons ($\sigma_\ell = 0.85$, clip [180, 1440] minutes). The configured means are targets of the unclipped variable, not of the clipped one.

Tier	Share	Target mean m	σ_ℓ	Clip (min)	S1 realised mean
normal	$\approx 82\%$	25 min	0.50	[5, 120]	25.0 min
surge	18%	90 min	0.65	S1 [30, 480]; S2 [30, 360]	90.1 / 90.0 min
dragon	32 bps / 25 bps	420 / 360 min	0.85 / 0.82	S1 [180, 1440]; S2 [200, 1440]	389.9 / 302.8 min

Table 4. Delay configuration (config.py) against realised means. Dragon realised means sit below the unclipped target because of the right clip at 24 hours and the left clip at 3–3.3 hours.

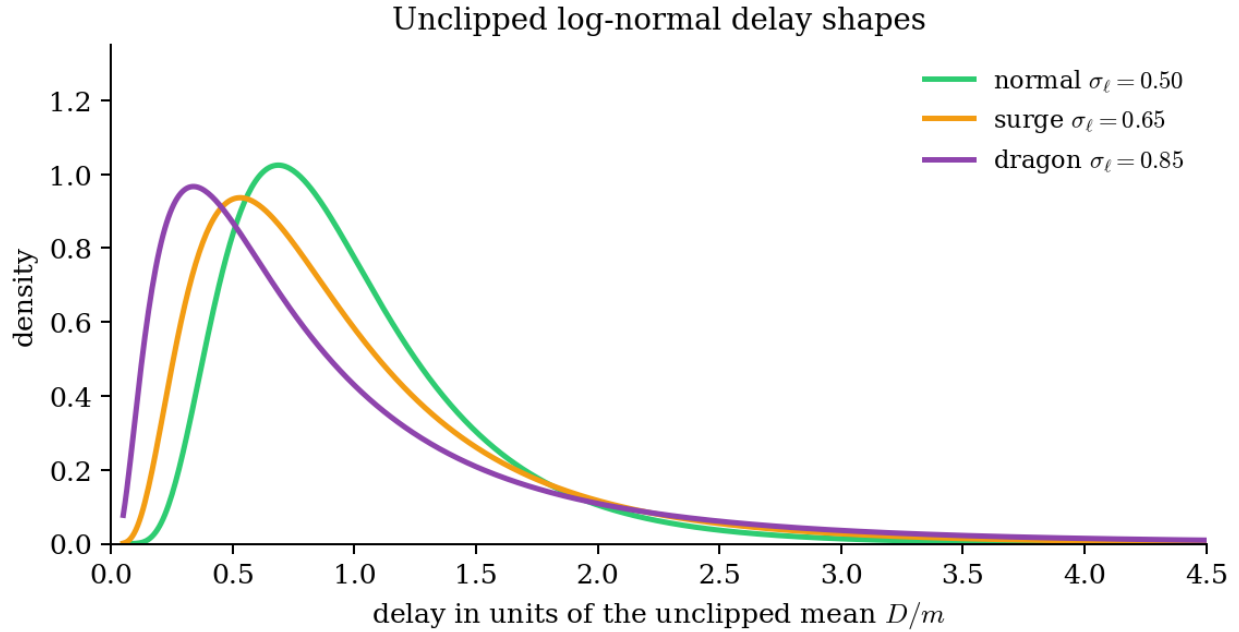


Figure 7. Shape of the unclipped log-normal for the three configured σ_ℓ values, plotted in units of the mean. Larger σ_ℓ is a heavier relative tail; clipping (not shown) then cuts both ends.

6.4 SLA bins and the unfulfilled-dragon indicator

SLA status is a deterministic function of delay, via `pd.cut(..., right=False)` on the bins $[0, 30, 120, 480, \infty)$ minutes in Scenario 1 and $[0, 30, 120, 360, \infty)$ in Scenario 2, labelled On-Time, Minor Delay, SLA Breach, Fulfilment Failure. Because the cuts are left-closed, a delay of exactly 30 minutes is Minor Delay, not On-Time. An unfulfilled dragon is a dragon whose delay also clears an explicit breach threshold τ (240 min = 4 h in S1; 360 min = 6 h in S2) and whose SLA label is one of the two breach labels:

$$U_i = 1\{T_i = \text{dragon}\} 1\{D_i \geq \tau\} 1\{\text{SLA}_i \in \{\text{Breach}, \text{Failure}\}\}. \quad (30)$$

On the recorded run this identifies 66 of 127 dragons (52.0%) in Scenario 1 and 17 of 66 (25.8%) in Scenario 2. The Scenario 2 rate is lower both because τ is higher (6 h vs 4 h) and because the dragon mean delay is lower (360 vs 420). Those two modelling choices are why Scenario 2 is the more conservative operational view, not any property of netting itself—although netting also removes the £168k ticket, which would otherwise dominate unfulfilled revenue.

6.5 Holding cost

Working-capital drag is charged at an annual holding rate $h = 0.25$. Delay in days is $D_i/1440$, so

$$H_i = V_i \cdot (D_i/1440) \cdot h. \quad (31)$$

This is a linear carrying-cost identity, not an inventory model: there is no (Q, r) policy, no cycle stock, no obsolescence. Summing over the file gives holding of £75,635.58 (0.849% of gross) in Scenario 1 and £59,274.45 (0.713%) in Scenario 2. Holding is a small drag. The loss that matters is the unfulfilled-dragon revenue $\sum_i U_i V_i$, equal to £147,582.95 (S1) and £31,158.62 (S2).

Remark. Two different “revenue at risk” numbers are printed in the delay stage. One is the sum of V_i over all tickets whose SLA is breach or failure, including non-dragons (£694,291 in S1, 7.791% of gross). The other is the unfulfilled-dragon sum (£147,583). The Monte Carlo, causal annual impact, and Hawkes P&L all use the dragon sum. The broader SLA sum is a dashboard statistic, not the loss functional of Sections 7–10. Section 13’s reconciliation table keeps them on separate rows.

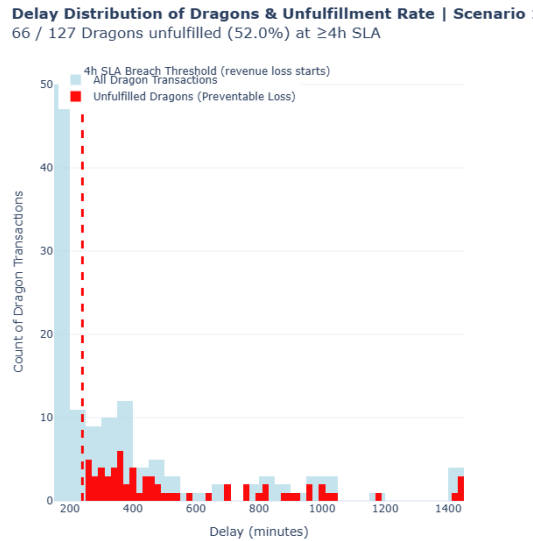


Figure 7b. Simulated delay distribution of dragon tickets, Scenario 1, with the 4-hour SLA threshold marked. Unfulfilled dragons are those to the right of the threshold.

7. Monte Carlo value-at-risk and expected shortfall

7.1 VaR and ES, as functionals

Definition (Value-at-risk and expected shortfall).

Let L be a loss random variable with distribution function F_L . For $\alpha \in (0,1)$ (here $\alpha = 0.95$ or 0.99), $\text{VaR}_\alpha(L) = \inf\{\ell : F_L(\ell) \geq \alpha\} = F_L^{-1}(\alpha)$. Expected shortfall is $\text{ES}_\alpha(L) = (1/(1-\alpha)) \int_\alpha^1 \text{VaR}_u(L) du$. If F is continuous, this equals $E[L \mid L \geq \text{VaR}_\alpha(L)]$. ES is coherent (Artzner et al., 1999); VaR is not, because it can penalise diversification. The pipeline’s headline is $\text{ES}_{0.95}$.

On a Monte Carlo sample $L^{(1)}, \dots, L^{(P)}$ of size $P = 10,000$, the estimators are the empirical quantile and the tail mean:

$$\widehat{\text{VaR}}_\alpha = L_{(\lceil \alpha P \rceil)}, \quad \widehat{\text{ES}}_\alpha = \frac{1}{\#\{p: L^{(p)} \geq \widehat{\text{VaR}}_\alpha\}} \sum_{p: L^{(p)} \geq \widehat{\text{VaR}}_\alpha} L^{(p)}. \quad (32)$$

That is the order-statistic convention. The code calls `numpy.percentile`, which linearly interpolates (Hyndman–Fan type 7). For $P = 10,000$ the two conventions differ by a fraction of a path. The expected-shortfall estimator is the sample mean of paths at or above that percentile, matching the code.

7.2 The simulated annual loss

Let V_d be average daily dragon revenue, computed as total dragon revenue divided by the number of calendar days in the sample (including days with no dragon):

$$V_d = \frac{1}{\Delta} \sum_{i \in D} V_i, \quad \Delta = (\max t - \min t) \cdot \text{days} + 1. \quad (33)$$

Let r_V be the revenue breach rate, the share of dragon revenue that sits on unfulfilled dragons, and let $g = 0.30$ be the gross-margin factor:

$$r_V = \frac{\sum_i U_i V_i}{\sum_{i \in D} V_i}, \quad g = 0.30. \quad (34)$$

Daily dragon revenue on a simulated day is taken log-normal with mean V_d and coefficient of variation equal to the CV of dragon-active days only, capped at 3:

$$\text{CV} = \min(3, s_{\text{active}} / \bar{R}_{\text{active}}). \quad (35)$$

Method of moments for a log-normal R with $E[R] = V_d$ and CV as above:

$$\sigma_d = \sqrt{\log(1 + \text{CV}^2)}, \quad \mu_d = \log V_d - \frac{1}{2} \log(1 + \text{CV}^2). \quad (36)$$

A path $p = 1, \dots, P$ is a year of such days, converted to margin-adjusted preventable loss by a constant factor:

$$L^{(p)} = g r_V \sum_{t=1}^{365} R_t^{(p)}, \quad R_t^{(p)} = \exp(\mu_d + \sigma_d Z_t^{(p)}). \quad (37)$$

Daily revenues are i.i.d. log-normal. Hawkes clustering is estimated on the same dragons in Section 9 and is not injected into these paths (no thinning, no stochastic intensity). That is the LDA reference construction: a marginal annual loss, not a clustered year. `TARGET_ANNUAL_DRAGONS` in `config.py` (104 / 72) is printed in this module and is not used in the path equation.

7.3 The mean of L is not the printed EL

Take expectations in the path equation. $E[R_t] = V_d = (\sum_D V_i)/\Delta$, so

$$E[L] = gr_V \cdot 365 \cdot V_d = g \cdot \frac{365}{\Delta} \sum_i U_i V_i. \quad (38)$$

The quantity the Monte Carlo module prints as “Expected Loss (EL) — data-derived” is the gross annualisation $(365/\Delta) \sum U_i V_i$ without g . Call it EL_{gross} . Then $E[L] = g \cdot EL_{\text{gross}}$. On Scenario 1, $EL_{\text{gross}} = £144,031$ and $g \cdot EL_{\text{gross}} = £43,209$, which is exactly the causal module’s “total annual impact” and sits next to the Monte Carlo median of £42,387 (right-skew: mean above median). The headline $ES_{0.95} = £60,134$ is a tail functional of L , i.e. already margin-adjusted. Comparing £144,031 to £60,134 as if they were estimates of the same thing is a category error; Section 13 tabulates this.

7.4 Unexpected loss

The printed “unexpected loss” is the gap between $Var_{0.95}$ and the median of L , $UL = Var_{0.95} - \text{median}(L)$, equal to £12,192 (S1) and £1,968 (S2). This is a dispersion measure of the simulated annual P&L, not a regulatory UL from a credit-risk IRB formula. It is small relative to EL_{gross} because the i.i.d. log-normal year-sum, with CV capped at 3, is not a disaster distribution once it has been averaged over 365 days. Clustering (Section 9) would fatten this UL if it were folded into the paths; it is not.

Functional of L (margin-adjusted unless noted)	Scenario 1	Scenario 2
$EL_{\text{gross}} = (365/\Delta) \sum U_i V_i$ (not a mean of L)	£144,031	£30,409
$g \cdot EL_{\text{gross}} = E[L]$ (approx.)	£43,209	£9,123
Median of L	£42,387	£9,021
$Var_{0.95}(L)$	£54,579	£10,989
$Var_{0.99}(L)$	£62,998	£12,062
$ES_{0.95}(L)$ (MC module headline)	£60,134	£11,716
$UL = Var_{0.95} - \text{median}$	£12,192	£1,968
Active-day CV (capped at 3)	3.00	2.28
Paths × horizon	10,000 × 365	10,000 × 365
Seed	314159	314159

Table 5. Monte Carlo module outputs. ES_{95} here is the headline of `src/risk/monte_carlo.py`. It is not the last-window backtest ES_{95} of Section 12.

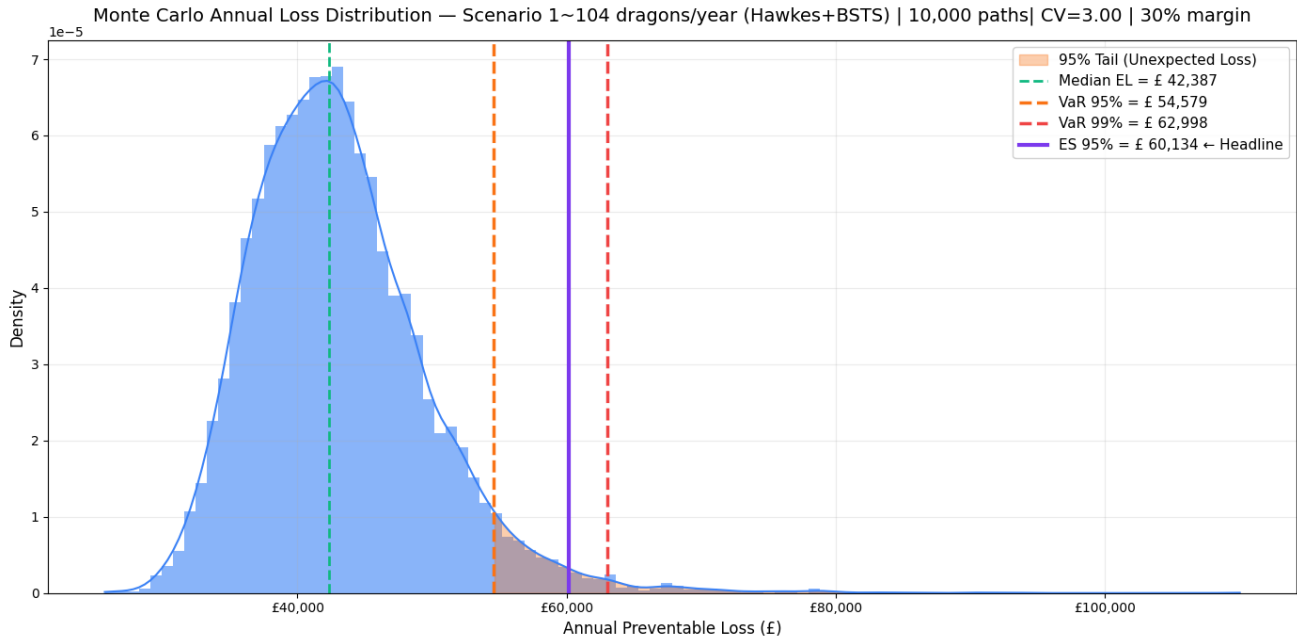


Figure 8. Monte Carlo annual loss density, Scenario 1, $P = 10,000$. Vertical lines are median, VaR95, VaR99 and ES95 of L (margin-adjusted). The paths are i.i.d. log-normal.

8. Causal contrasts: ATE as negative control, QTE as the object of interest

8.1 What is being identified

There are two binary flags, and they must not be conflated. The surge flag $W_i = 1\{i \in S\}$ is assigned uniformly among non-dragons. The dragon flag $\delta_i = 1\{i \in D\}$ is assigned with weights V_i^Y . The causal engine's propensity-score and quantile-regression machinery treats W as a treatment and net revenue R^{net} as an outcome. The dragon machinery treats δ as a grouping variable and V as an outcome. Only the second grouping is economically interesting, and it is not a treatment in the Neyman–Rubin sense.

8.2 Why the surge ATE must be near zero

Potential outcomes $(R^{\text{net}}(1), R^{\text{net}}(0))$ for surge would be well-defined if surge were an intervention that changed delay and, through delay, holding cost. In the simulator that is true in Scenario 1: W_i changes the delay mean from 25 min to 90 min, so H_i rises by $V_i \cdot (65/1440) \cdot 0.25 \approx 0.011 V_i$, about 110 basis points of value. The average treatment effect on net revenue is therefore on the order of minus one percent of average ticket value, i.e. a few tens of pence, plus noise from matching. There is no path from W to V , because V is determined before assignment. Empirically the caliper-matched ATE is £7 (S1) and –£21 (S2), and quantile regression coefficients on W are ≈ 0 at the configured quantiles $\{0.95, 0.999\}$, with a –£10 coefficient at the 99.9th in Scenario 1. That is a passed negative control, not a finding about warehouse physics.

8.3 Propensity scores

Rosenbaum and Rubin (1983): if unconfoundedness $Y(0), Y(1) \perp W \mid X$ and overlap $0 < P(W=1|X) < 1$ hold, then $Y(0), Y(1) \perp W \mid \pi(X)$ where $\pi(x) = P(W=1 \mid X=x)$. A logit model

$$\text{logit } \pi(x) = x^\top \beta \quad (39)$$

is fitted by L2-penalised maximum likelihood (logistic regression, saga solver, $C=1$) on a without-replacement subsample of 50,000 tickets, after building a treated-plus-four-to-one control pool. Scores are then predicted for every row. Covariates are country, log quantity and hour of the invoice timestamp. Matching with caliper $c = 0.005$ keeps a treated unit t and a control u only if $|\pi_t - \pi_u| \leq c$, within country. Scenario 1 forms a within-bin Cartesian join and keeps, for each treated propensity, the nearest control (a control may match more than one treated unit). Scenario 2 walks sorted propensity scores with a used-flag and is one-to-one without replacement. The matching ATE is

$$\widehat{\text{ATE}}_{\text{PSM}} = \frac{1}{M} \sum_{m=1}^M (R_{t(m)}^{\text{net}} - R_{c(m)}^{\text{net}}). \quad (40)$$

8.4 Quantile regression

Koenker and Bassett (1978). For $q \in (0,1)$ the check function $\rho_q(u) = u(q - 1\{u < 0\})$ yields

$$\hat{\beta}(q) = \min_{\beta} \sum_i \rho_q(R_i^{\text{net}} - x_i^\top \beta). \quad (41)$$

The coefficient on W at quantile q is a q -quantile treatment effect under a linear model for the conditional quantile of net revenue. Fitted at $q \in \{0.95, 0.999\}$ on a subsample of 180,000. All estimated coefficients are ≈ 0 to the printed precision, except $q = 0.999$ in Scenario 1 ($-\pounds 10$). The check-function asymptotics are delicate in the far tail, the subsample is not clustered by day, and the true ATE is a holding-cost effect of pennies to pounds.

8.5 Quantile treatment effects of the dragon label

Definition (QTE of dragon on value).

The q -quantile treatment effect of the dragon grouping on order value is $\Delta(q) = F_{\{V|\delta=1\}^{-1}}(q) - F_{\{V|\delta=0\}^{-1}}(q)$. It is estimated by the difference of sample quantiles. It is a descriptive contrast, not an ATE of a manipulable treatment, because δ is a function of V .

The dragon premium is the mean analogue, $\pi = E[V | \delta=1] - E[V | \delta=0]$. On the recorded run, with the configured quantile list $\{0.95, 0.999\}$:

Contrast	Scenario 1	Scenario 2
Dragon mean premium	£2,837	£1,700
QTE 95%	£3,675	£7,665
QTE 99.9%	£156,138	£12,069
PSM ATE of surge on net revenue	£7	−£21

Contrast	Scenario 1	Scenario 2
QR coefficient on surge, $q=0.95$	£0	£0
QR coefficient on surge, $q=0.999$	-£10	£4

Table 6. Causal-engine outputs. The surge ATE/QR block is the negative control. The dragon premium and QTE block is the object of interest, and is partly mechanical from value-biased sampling. Current `config.py` keeps `QUANTILES = [0.95, 0.999]`.

The QTE profile in Scenario 1 is the picture the project is built to show. At the 95th the dragon ticket is £3.7k richer than a non-dragon; at the 99.9th, £156k. That is a regularly varying value tail, flagged by construction, and it is why a 4-hour SLA miss on a dragon is an entirely different economic event from a 4-hour miss on a median ticket. Scenario 2's QTE profile is flatter in the far tail because netting has removed the £168k PAPER CRAFT observation.

8.6 Annual impact

The causal module's "total annual impact" is not an ATE. It is the margin-adjusted, calendar-annualised unfulfilled-dragon revenue,

$$I = g \cdot \frac{365}{\Delta} \sum_i U_i V_i, \quad (42)$$

equal to £43,209 (S1) and £9,123 (S2). This is the same identity as $E[L]$ in Section 7.3. The causal engine does not add a cancellation-loss term (`will_Cancel` is hard-coded to 0). The dashboard's "annual dragons" is $\text{int}(|D| \cdot 365/\Delta)$, giving 123 and 64 respectively.

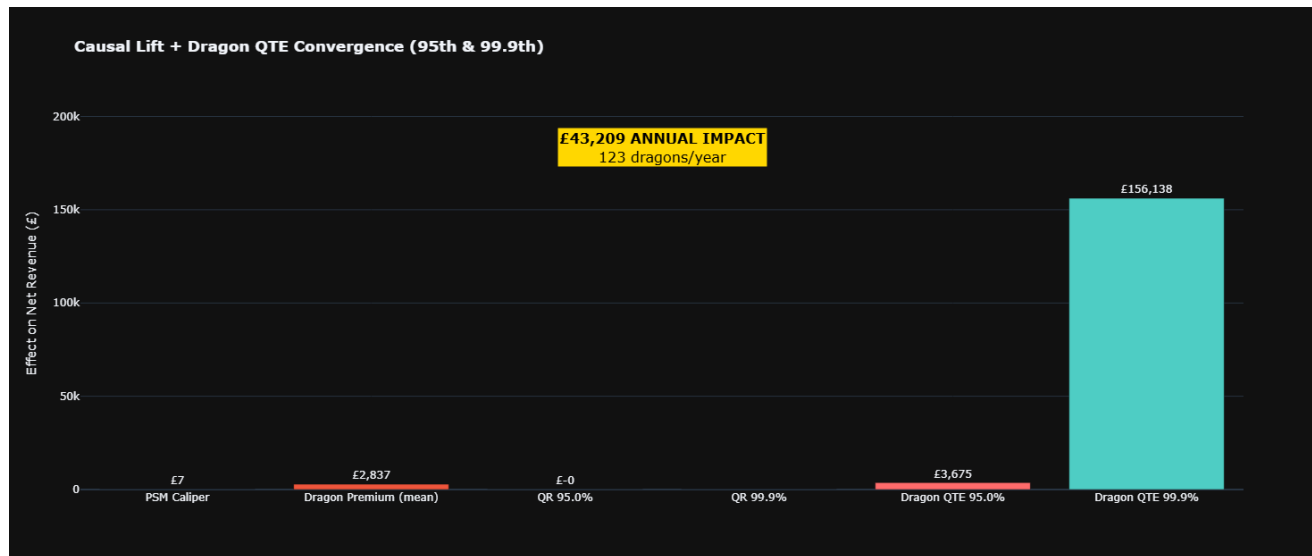


Figure 9. Scenario 1 causal panel: surge ATE/QR (negative control, near zero) against dragon premium and QTEs (the object of interest).

9. Hawkes self-exciting point processes

9.1 Intensity, compensator, branching ratio

Let $0 < t_1 < \dots < t_N \leq T$ be the dragon timestamps, measured in minutes from the first dragon. A (univariate) Hawkes process with exponential kernel has conditional intensity

$$\lambda(t) = \mu + \alpha \sum_{t_j < t} e^{-\beta(t-t_j)}. \quad (43)$$

Each event jumps the intensity by α and the jump decays at rate β . The process is a linear branching process in continuous time: immigrants arrive at rate μ , and each event produces offspring according to the kernel. The expected number of direct offspring of one event—the branching ratio—is

$$n = \int_0^\infty \alpha e^{-\beta s} ds = \alpha / \beta. \quad (44)$$

Stationarity in the Hawkes sense requires $n < 1$. If $n \geq 1$ the intensity explodes (subcritical vs critical branching). The half-life of a jump is $\log 2 / \beta$. The code rejects parameter candidates with $\mu \leq 0$, $\alpha < 0$, $\beta \leq 0$ or $(\alpha/\beta) \geq 1$, which is exactly this stationarity constraint.

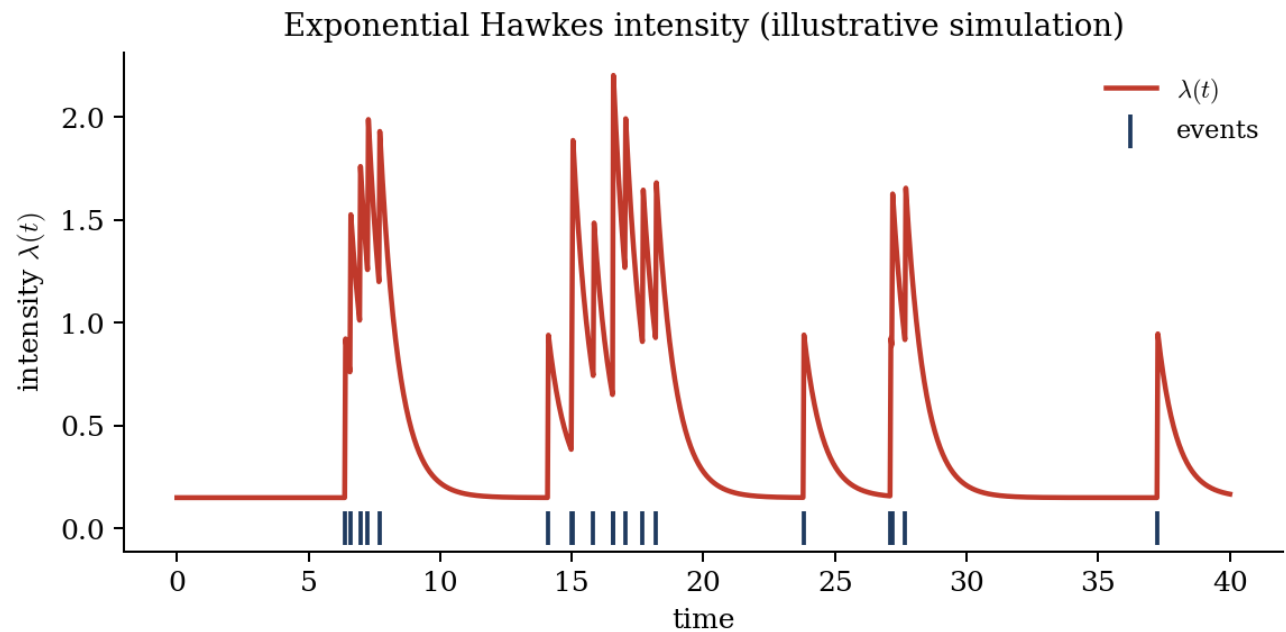


Figure 10. A simulated exponential Hawkes intensity (illustrative, not the warehouse fit): each event produces a jump that decays, and clustered events produce bursts.

9.2 Likelihood, and the implemented recursion

For a simple point process on $[0, T]$ observed as $t_1, \dots, t_N$, the log-likelihood (Daley and Vere-Jones; Ogata 1978) is

$$\ell(\mu, \alpha, \beta) = \sum_{i=1}^N \log \lambda(t_i) - \int_0^T \lambda(s) ds. \quad (45)$$

The integral is the compensator $\Lambda(T) = E[N(T) \mid \text{history}]$, and for the exponential kernel it has the closed form

$$\Lambda(T) = \mu T + \frac{\alpha}{\beta} \sum_{i=1}^N (1 - e^{-\beta(T-t_i)}). \quad (46)$$

The sum over $\log \lambda(t_i)$ is $O(N^2)$ if written naïvely. A recursion reduces it to $O(N)$. The implementation used here follows an *undecayed-jump* update. Set $A_1 = 0$. At event i the intensity contribution of the history is A_i , so $\log \lambda(t_i) = \log(\mu + A_i)$. Between events the accumulator is updated as

$$A_{i+1} = \alpha + A_i e^{-\beta \Delta t_i}, \quad \Delta t_i = t_{i+1} - t_i, \quad (47)$$

for $i < N - 1$. Each event therefore contributes an undecayed jump of size α into the intensity evaluated at the next event, after the previous accumulator has been decayed by $e^{-\beta \Delta t}$. The compensator uses (46) with observation horizon $T = t_N$, the last dragon timestamp. This is the recursion the MLE maximises; it is left as implemented.

The negative log-likelihood is passed to L-BFGS-B with bounds $\mu \geq 10^{-8}$, $\alpha \in [0, 0.99]$, $\beta \geq 0.1$, starting at $(2.5 \times 10^{-4}, 0.75, 1.2)$. On the recorded run the MLE sits at the α box bound in both scenarios, with a large decay ($\beta \approx 410 \text{ min}^{-1}$) and a small baseline ($\mu \approx 2 \times 10^{-6}$ per minute). The implied branching ratio is $n = \alpha/\beta \approx 0.0024$, and the half-life is $\log 2 / \beta \approx 0.10$ seconds.

9.3 Thirty-day expected counts

The 30-day forecast is the compensator increment from the last event,

$$E[N(t_N + \tau) - N(t_N)] = \int_{t_N}^{t_N + \tau} \lambda(s) ds, \quad (48)$$

with $\tau = 30 \times 1440$ minutes, evaluated by the trapezoid rule on 15,000 grid points of a Numba-parallel intensity that uses the exponential kernel (43) on the observed history. The printed value is 1.5 dragons in both scenarios.

Hawkes quantity	Scenario 1	Scenario 2
N (dragons)	127	66
μ (per minute)	1.9×10^{-6}	2.0×10^{-6}
α (jump)	0.990 (bound)	0.990 (bound)

Hawkes quantity	Scenario 1	Scenario 2
β (per minute)	410.349	405.170
Branching ratio α/β	0.0024	0.0024
Half-life $\log 2 / \beta$	0.10 s	0.10 s
Printed 30-day $E[N]$	1.5	1.5

Table 7. Fitted Hawkes parameters under the implemented undecayed-jump recursion. The interesting number is the branching ratio $n = \alpha/\beta$.

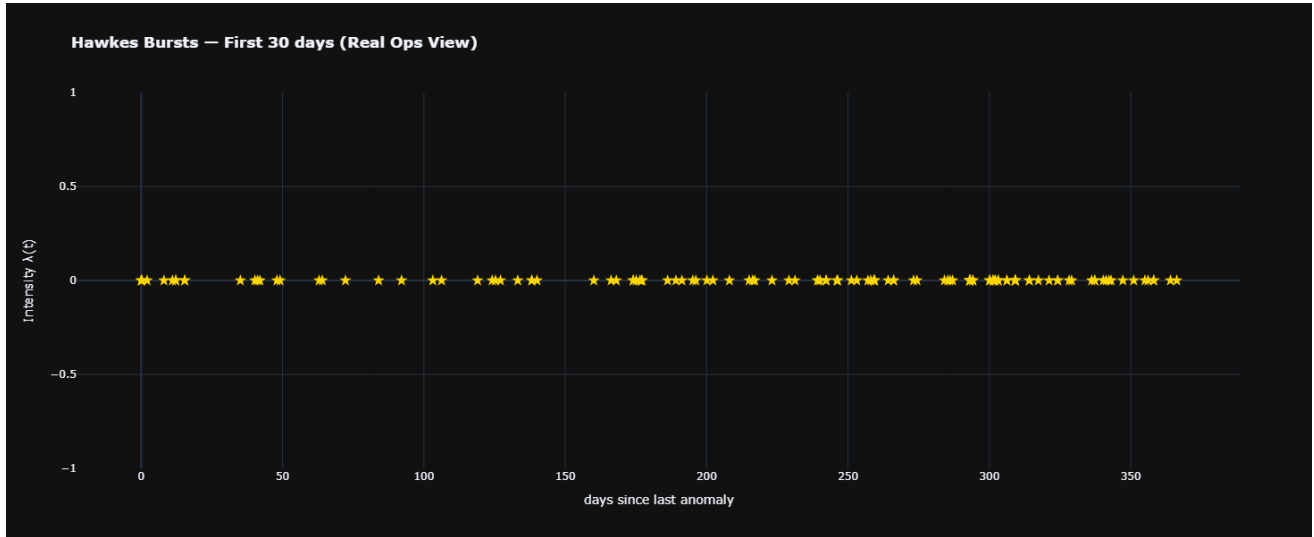


Figure 11. Dashboard Hawkes plot, Scenario 1. Intensity is evaluated on a grid starting at the last dragon. Gold markers are event times.

10. State-space projection of monthly dragon counts (Kalman local linear trend)

10.1 Local linear trend

What is implemented is a linear-Gaussian state-space model updated by the Kalman filter, with Q and R treated as fixed numbers. That is Harvey's structural time series in its filtering form. Let y_t be the dragon count in month t , $t = 1, \dots, m$, with missing months filled by zero. A local linear trend is

$$\mu_t = \mu_{t-1} + \beta_{t-1} + \eta_t, \quad \eta_t \sim N(0, q_\mu), \quad (49)$$

$$\beta_t = \beta_{t-1} + \zeta_t, \quad \zeta_t \sim N(0, q_\beta), \quad (50)$$

with process noises independent of the observation noise, $Q = \text{diag}(8 \times 10^{-4}, 5 \times 10^{-7})$ on the trend block, and $R = \text{numpy.var}$ of the strictly positive y ($1/n$ population variance of those months). In state-space form, $x_t = (\mu_t, \beta_t)^\top$,

$$x_t = Fx_{t-1} + w_t, \quad y_t = Hx_t + \varepsilon_t, \quad (51)$$

F is the 2×2 local-linear-trend matrix with $F_{11}=F_{12}=F_{22}=1$ and $F_{21}=0$. That is what the code realises by starting from an identity and setting $F[0,1] = 1$. The observation is $y_t = x_t[0]$, i.e. $H = (1, 0, \dots)$.

10.2 Kalman filter

Given a prior $x_{t|t-1}$, $P_{t|t-1}$, the prediction and update (Kalman, 1960) are

$$x_{t|t-1} = Fx_{t-1|t-1}, \quad P_{t|t-1} = FP_{t-1|t-1}F^\top + Q, \quad (52)$$

$$S_t = HP_{t|t-1}H^\top + R, \quad K_t = P_{t|t-1}H^\top S_t^{-1}, \quad (53)$$

$$x_{t|t} = x_{t|t-1} + K_t(y_t - Hx_{t|t-1}), \quad P_{t|t} = P_{t|t-1} - K_t S_t K_t^\top. \quad (54)$$

With H looking at coordinate 0 one has $S_t = (P_{t|t-1})_{00} + R$ and $K_t = (P_{t|t-1} \text{ column } 0)/S_t$, which is the scalar form in the code. Initialisation is $x_0[0] = \bar{y}$, $P_0 = 50 I$. The in-sample fit plotted in the dashboard is the sequence of one-step predictions $Hx_{t|t-1}$.

10.3 Projection, and the clip to {6,...,10}

Open-loop projection is $x_{m+k} = F^k x_{m|m}$, $\hat{y}_{m+k} = (x_{m+k})_0$. The printed monthly rate is not the raw open-loop path. Let $\bar{y}_{10:12}$ be the mean of the last three projected months, and let r be its nearest integer (exactly `int(np.round(np.mean(proj[-3:])))`). The code then projects that integer onto $\{6, \dots, 10\}$:

$$\tilde{y} = \min(10, \max(6, r)), \quad (55)$$

then annualises as $12 \tilde{y}$. Scenario 1 hits the cap (10/month $\rightarrow$ 120/year); Scenario 2 hits the floor (6/month $\rightarrow$ 72/year). This is the annualisation map used in the module: a clipped integer in a five-point set, then multiplied by twelve. `TARGET_ANNUAL_DRAGONS` in `config.py` is 104 (S1) and 72 (S2).

10.4 P&L bleed identity

Write N for the clipped annual count, $n_U = \sum U_i n_D = |D|$, $U = \sum U_i V_i$ (read from the unfulfilled-dragon frame). The anomaly premium used in this module is U/n_U (unfulfilled-dragon mean value), the breach rate is $n_U/$

n_D , and

$$B = g \cdot N \cdot \frac{U}{n_U} \cdot \frac{n_U}{n_D} = g \cdot N \cdot U / n_D. \quad (56)$$

Relative to the causal annual impact $I = g (365/\Delta) U$, the bleed replaces the empirical annualisation $365/\Delta \cdot n_D$ by the clipped Kalman count N . That is the only structural difference. On the recorded run, $B = \text{£}41,835$ (S1, $N = 120$) and $\text{£}10,197$ (S2, $N = 72$).

10.5 Mitigation multipliers

Four scenario multipliers $\rho \in \{0.70, 0.55, 0.15, 0.00\}$ are applied to N , and savings are $B(1-\rho)$. They are not estimated treatment effects of rerouting or safety stock. They are proportional-reduction arithmetic, labelled with operational names. They should be read as a sensitivity table on N , not as an operations-research optimisation of safety stock. Q and R are not estimated (no EM, no Gaussian MLE, no Bayes); with $m \approx 13$ monthly points, fixing Q , R and clipping $\tilde{y}$ is the computational map the module uses.

Quantity	Scenario 1	Scenario 2
Clipped monthly rate $\tilde{y}$	10 (cap)	6 (floor)
Annual $N = 12 \tilde{y}$	120	72
Bleed $B = g N U / n_D$	£41,835	£10,197
5-year arithmetic 5B	£209,173	£50,987
Reroute 30% ($\rho=0.70$), saving	£12,550	£3,059
Safety stock 7 SKUs ($\rho=0.55$), saving	£18,826	£4,589
Full mitigation ($\rho=0.15$), saving	£35,559	£8,668
Zero anomalies ($\rho=0$), saving	£41,835	£10,197

Table 8. Kalman-clipped count forecasts and proportional mitigation arithmetic. Savings = $B(1-\rho)$.

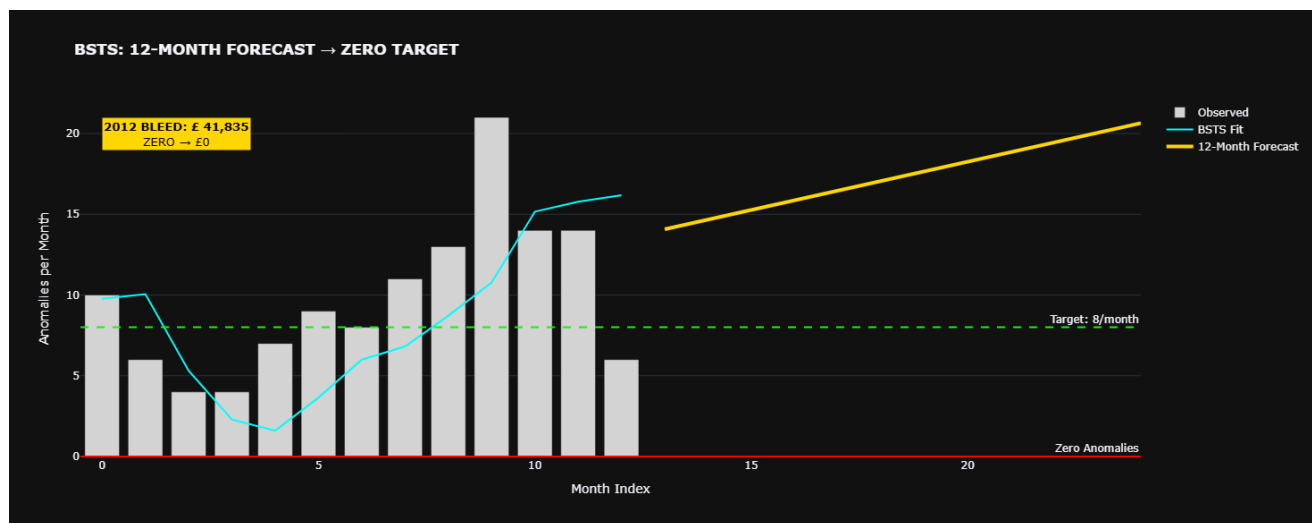


Figure 12. Kalman local-linear-trend fit and 12-month open-loop projection, Scenario 1. The printed 2012 count is the clipped monthly rate times 12, not the raw open-loop path.

11. Peaks-over-threshold GPD on daily realised loss

Section 5 fitted a tail to quantity. The backtest module fits a GPD to a different random variable: daily realised loss. Define, on each ticket,

$$L_i = U_i (V_i + H_i), \quad (57)$$

and aggregate $\ell_t = \sum_{i: \text{day}(i)=t} L_i$. Most days have $\ell_t = 0$. The 99th percentile u of $\{\ell_t\}$ is the POT threshold (GPD_THRESHOLD_PCT = 99.0). Excesses $Y_j = \ell_t - u$ on $\{\ell_t > u\}$ are modelled as GPD(ξ, β), with MLE as in Section 5.10.

11.1 VaR and ES of the parent, from a GPD excess

Let n_{obs} be the number of distinct invoice dates in the daily-loss series and N_u the number of exceedances. For $x > u$ the POT approximation is

$$P(\ell > x) \approx \frac{N_u}{n_{\text{obs}}} (1 + \xi(x - u)/\beta)^{-1/\xi}. \quad (58)$$

Value-at-risk at tail probability α (the code uses $\alpha = 0.05$, i.e. a 95% query of the daily-loss distribution) solves $P(\ell > \text{VaR}) = \alpha$, hence

$$\text{VaR}_{1-\alpha} = u + \frac{\beta}{\xi} ((\alpha n_{\text{obs}} / N_u)^{-\xi} - 1) \quad (59)$$

for $\xi \neq 0$, and $\text{VaR} = u - \beta \log(n_{\text{obs}} \alpha / N_u)$ for $\xi = 0$. The code also enforces $\text{VaR} \geq u$. That clamp is part of the method. The GPD is fitted above the 99th percentile, so $N_u / n_{\text{obs}} \approx 0.01$, while the query is a 95% VaR ($\alpha = 0.05$). The unconstrained 95% quantile lies below u , and the clamp therefore sets $\text{VaR}_{0.95} = u$. Expected shortfall of a GPD parent has the closed form (McNeil, Frey and Embrechts)

$$\text{ES}_{1-\alpha} = \frac{\text{VaR}_{1-\alpha} + \beta - \xi u}{1 - \xi}, \quad (60)$$

which is the line $(\text{var95_gpd} + \text{beta} - \text{xi} * u) / (1 - \text{xi})$. For $\xi \rightarrow 0$ this collapses to $\text{VaR} + \beta$, the exponential tail mean. Substituting the clamped $\text{VaR}_{0.95} = u$ gives $\text{ES}_{0.95} = u + \beta / (1 - \xi)$, which is $E[\ell \mid \ell > u]$ — the GPD mean above the 99th percentile. Check: $3,990 + 30,440.76 / 0.952 \approx 35,965$, matching the printed £35,966. The pair is therefore a far-tail daily exhibit: how bad a day above the 99th-percentile loss threshold is, labelled at the 95% query because that is the α the module asks for.

11.2 Fitted numbers

GPD on daily realised loss	Scenario 1	Scenario 2
Threshold u (99th pctile of ℓ_t)	£3,990	£1,745
Exceedances N_u	4	4
ξ	+0.0480	-0.0881
β	30,440.76	5,829.39
VaR _{0.95} (daily)	£3,990	£1,745
ES _{0.95} (daily)	£35,966	£7,102
Domain of daily loss	weakly Fréchet	Weibull (bounded)

Table 9. POT GPD fitted to daily realised loss in the backtest module. These are daily, not annual, functionals. Printed VaR95 equals u because $\alpha = 0.05$ lies below the 99th-percentile exceedance rate.

$N_u = 4$ in each scenario is a four-point fit of a two-parameter model; a two-parameter MLE on four excesses, with ξ bounded above by 1, will sit near the exponential ($\xi \approx 0$), which is what 0.048 is. Scenario 2's $\xi = -0.088$ is a Weibull-domain daily-loss tail: after netting, daily unfulfilled-dragon loss is a rare, bounded-looking compound, even though quantity itself is Fréchet. There is no contradiction. Quantity and daily loss are different variables, and the daily-loss series is mostly zeros.

12. Purged expanding-window backtesting

12.1 The window algebra

The backtest module is a replay of the delay DGP (same seed 314159), not a continuation of the delay-stage frame. That is why its coverage table is internally consistent with the printed engine. Let the sorted unique dates be $d_1, \dots, d_T$. An expanding window is indexed by i , starting at the first date at least MIN_TRAIN_DAYS = 90 after d_1 , stepping by WINDOW_STEP = 7 dates, with a purge of PURGE_GAP = 3 days:

$$\text{train} = \{d \leq d_i - 3\}, \quad \text{test} = \{d_i - 3 < d \leq d_i\}. \quad (61)$$

Windows with fewer than 50,000 training tickets or fewer than 5 test tickets are skipped. The purge exists so that the test slice does not share invoices, information, or overlapping delay draws with the end of train—the same reason López de Prado purges embargoed observations in combinatorial CV. On this file one obtains 34 scored windows in Scenario 1 and 33 in Scenario 2. The calendar span used to annualise gross is $(\text{max date} - \text{min date}).\text{days} + 1$.

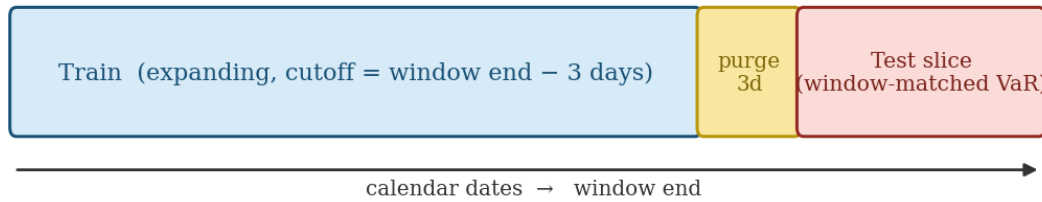


Figure 13. One expanding-window step as implemented: train up to the cutoff, a three-day purge, a short test slice up to the window end. The violation test compares test realised loss to a window-matched Monte Carlo VaR, not to the annual VaR.

12.2 Two Monte Carlo scales inside one loop

On each training slice the module re-estimates the log-normal of Section 7, with one change: the mean daily dragon revenue is the mean over dragon-active training days, not over all calendar days. Paths of length 365 give an annual loss distribution; its 95% quantile and tail mean are the rolling annual VaR95/ES95 plotted in the dashboard. Independently, the first d_{test} columns of the same path matrix (typically 3 days) are summed to a window-matched loss; the 95% quantile of that short-horizon distribution is the VaR against which the test realised loss is scored. A violation is

$$I_k = 1 \left\{ \sum_{i \in \text{test}_k} L_i > \widehat{\text{VaR}}_{0.95}^{\text{window}, k} \right\}. \quad (62)$$

This scale-matching is the right idea: one must not test a three-day realised loss against an annual VaR. Test realised loss is $\Sigma (V_i + H_i)$ over unfulfilled dragons in the slice (gross, no margin). Window VaR is a quantile of $g r_V \Sigma R_t$ (margin-adjusted dragon revenue). A violation is therefore a comparison of two different currencies of loss. That does not make the test useless; it does mean a 0/34 violation rate is not a calibration theorem.

12.3 Kupiec's unconditional coverage test

Under a correctly calibrated VaR_{1-p} with $p = 0.05$, the indicators I_k are Bernoulli(p) (and, if the windows do not overlap in information, independent). Let n_w be the number of windows and k the number of violations. Kupiec's proportion-of-failures likelihood-ratio test of $H_0: E[I] = p$ is

$$\text{LR}_{\text{uc}} = -2 \log \frac{(1-p)^{n_w-k} p^k}{(1-\hat{\pi})^{n_w-k} \hat{\pi}^k}, \quad \hat{\pi} = k/n_w, \quad (63)$$

with p -value $1 - F_{\chi^2_1}(\text{LR}_{\text{uc}})$. At the boundary the code uses the limiting forms: if $k = 0$, $\text{LR} = -2 n_w \log(1-p)$; if $k = n_w$, $\text{LR} = -2 n_w \log p$. For Scenario 1, $n_w = 34$ and $k = 0$, this is $\text{LR} \approx 3.49$, p -value ≈ 0.062 — a borderline non-rejection at 5%. Expected violations under $p = 0.05$ are about 1.7 in 34 windows: zero is unsurprising, and power is low. Scenario 2, with 2/33 violations (6.1%), has a well-defined interior LR and Kupiec $p = 0.7864$.

12.4 Christoffersen's independence test

Independence of exceptions is tested as a first-order Markov chain against i.i.d. Bernoulli. Count transitions $n_{ij} = \#\{k : I_{k-1}=i, I_k=j\}$ for $i,j \in \{0,1\}$, write $\pi_{01} = n_{01}/(n_{00}+n_{01})$, $\pi_{11} = n_{11}/(n_{10}+n_{11})$, $\pi = (n_{01}+n_{11})/n'$, and

$$\text{LR}_{\text{ind}} = -2(\ell_{\text{i. i. d.}} - \ell_{\text{Markov}}). \quad (64)$$

The restricted likelihood is that of i.i.d. Bernoulli(π); the unrestricted is the two-probability Markov chain. Transition probabilities are clipped away from $\{0,1\}$ for numerical stability. Christoffersen's conditional coverage test (not computed here) would be $\text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}} \sim \chi^2_2$. Scenario 2 reports Christoffersen $p = 0.6054$: no evidence of clustered exceptions in a sample of 33 indicators, which is also a statement about power.

12.5 Last-window Monte Carlo, the other ES95

The P&L block of the backtest prints a second pair (VaR95_{MC} , ES95_{MC}) taken from the last expanding window's 365-day paths. Because that window uses the active-day mean, the numbers are much larger than Section 7: $\text{ES95}_{\text{MC}} = \text{£}248,220$ (S1) and $\text{£}75,001$ (S2) on the recorded run. They are a different functional of a different mean. They are not a revision of $\text{£}60,134$ / $\text{£}11,716$. Appendix A.5 writes the $1/f$ scaling that produces the gap.

Backtest quantity	Scenario 1	Scenario 2
Windows scored	34	33
Violations	0 (0.0%)	2 (6.1%)
Kupiec p (implemented LR)	≈ 0.062 (k=0 form)	0.7864
Christoffersen p	1.0000	0.6054
Last-window MC VaR95 (annual)	£224,246	£69,656
Last-window MC ES95 (annual)	£248,220	£75,001
GPD daily ES95	£35,966	£7,102

Table 10. Purged expanding-window backtest, as implemented (90 / 7 / 3). Kupiec at $k = 0$ uses $\text{LR} = -2 n_w \log(1-p)$.

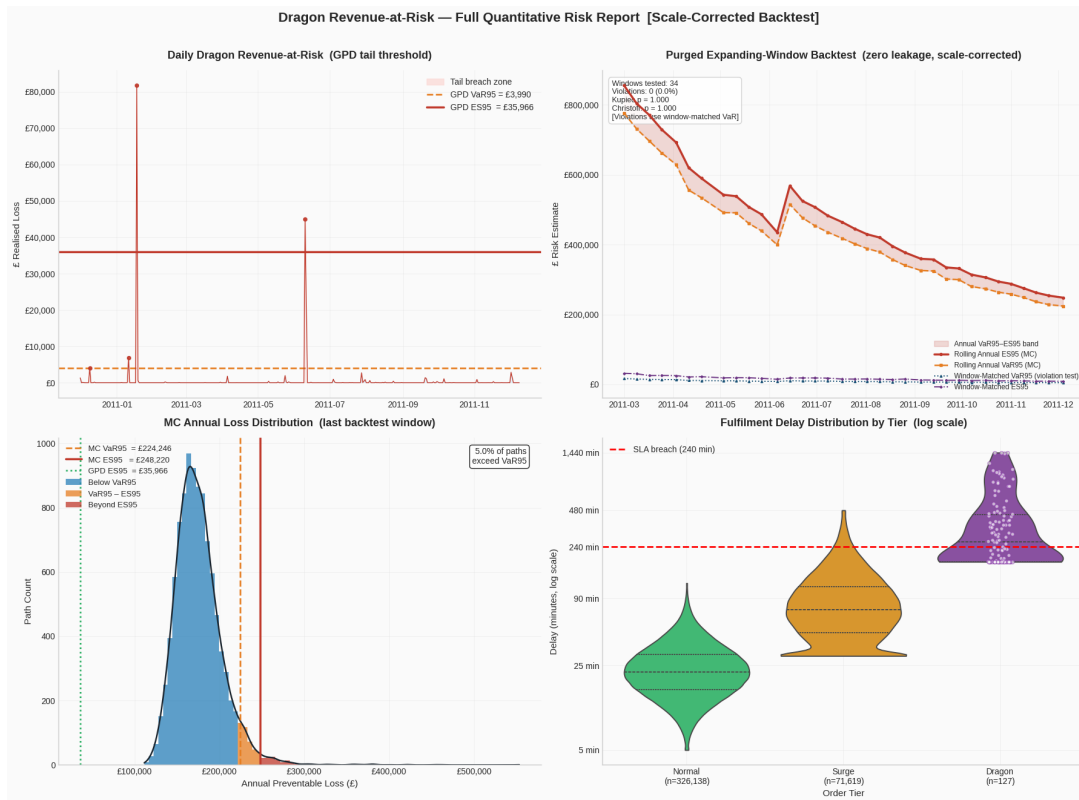


Figure 14. Four-panel quantitative risk report, Scenario 1: daily loss with GPD thresholds; rolling annual and window-matched VaR/ES; last-window MC histogram; log-delay violins by tier.

13. Reconciliation of every reported sterling figure

The repository prints several loss numbers per scenario. They look like rivals. They are not. Each is a different functional, of a different random variable, sometimes with and sometimes without the 30% margin, sometimes using all calendar days and sometimes only dragon-active days, sometimes annual and sometimes daily. This section writes the identity for each number, plugs in the recorded run, and states what question that number actually answers.

13.1 The primitives, Scenario 1

On the recorded gross run the following primitives are fixed by the delay stage (seed 314159): $n = 397,884$ tickets; $\Delta = 374$ days; $\Sigma V_i = £8,911,407.90$; $|D| = 127$ dragons; $U = \Sigma U_i V_i = £147,582.95$; $n_U = 66$; $g = 0.30$; $h = 0.25$; $\Sigma H_i = £75,635.58$; broader SLA-breach revenue (all tiers) = £694,291.

13.2 The identities

(A) Holding drag, not a tail functional. $\Sigma H_i = \Sigma V_i (D_i/1440) h = £75,635.58$. Question answered: how much working-capital cost does the simulated delay clock charge, on every ticket.

(B) Broad SLA revenue-at-risk. $\Sigma \{V_i : \text{SLA}_i \in \{\text{breach, failure}\}\} = £694,291$, annualised $\times 365/\Delta$ to £677,583. Question answered: value sitting on any SLA miss, dragons or not. Not used downstream.

(C) Unfulfilled-dragon revenue, sample. $U = \Sigma U_i V_i = £147,583$. Question answered: value sitting on dragons that cleared the 4-hour threshold in this 374-day file.

$$\text{EL}_{\text{gross}} = \frac{365}{\Delta} U. \quad (65)$$

(D) Data-derived EL_{gross} is (C), stretched to a 365-day year, still gross of margin. This is the Monte Carlo module's "EL".

$$I = g \cdot \frac{365}{\Delta} U = E[L]. \quad (66)$$

(E) Margin-adjusted annual impact, also $E[L]$ of the i.i.d. MC. This is the causal dashboard's "total annual impact", and it is the mean of the Monte Carlo loss L of Section 7, up to Monte Carlo error and the CV-cap.

(F) Monte Carlo median, VaR, ES of L . $L^{(p)} = g r_V \sum_{t=1}^{365} R_t^{(p)}$ with R_t log-normal of all-day mean and active-day CV. Median £42,387, VaR95 £54,579, VaR99 £62,998, ES95 £60,134. Question answered: the distribution of one-year margin-adjusted preventable loss under i.i.d. daily dragon revenue. The headline of `monte_carlo.py`.

$$B = g \cdot N \cdot U / n_D. \quad (67)$$

(G) Kalman/Hawkes bleed. Question answered: (E), but with the empirical annualisation $365/\Delta \cdot n_D$ replaced by the clipped Kalman count $N = 120$ (S1) or 72 (S2).

(H) Last-window backtest MC ES95. Same path equation as (F), but $E[R]$ is the mean of dragon-active days in the last training slice, not V_d over all calendar days, and the training slice is not the full sample. ES95 = £248,220 (S1), £75,001 (S2). Question answered: a more aggressive annual loss, conditional on "a dragon day looks like a dragon day".

(I) GPD daily ES95. Printed as ES of the daily realised-loss series under a POT GPD at $\alpha = 0.05$: £35,966 (S1), £7,102 (S2). Because the GPD is fitted above the 99th percentile and VaR is clamped to u , this is $E[\ell \mid \ell > u_{99}]$ ($V+H$, no margin). Not annualised.

(J) "Tail scenario daily $\times 252$ ". 99th percentile of daily realised loss, times 252: £1,005,366 (S1), £439,753 (S2). Question answered: a crude annualisation of a bad day as if it occurred every trading day. It is an upper-stress exhibit, not an expectation, and it should not be added to (F) or (H).

13.3 Master reconciliation table

ID	What is printed	S1	S2	$\times g?$	Horizon
A	Holding cost ΣH	£75,636	£59,274	n/a	sample
B	Broad SLA RaR	£694,291	£463,771	no	sample
C	Unfulfilled-dragon U	£147,583	£31,159	no	sample
D	$\text{EL}_{\text{gross}} = (365/\Delta)U$	£144,031	£30,409	no	year

ID	What is printed	S1	S2	× g?	Horizon
E	Causal annual impact $g(365/\Delta)U$	£43,209	£9,123	yes	year
F	MC module ES95 of L	£60,134	£11,716	yes	year
F'	MC module VaR95 of L	£54,579	£10,989	yes	year
G	Kalman bleed $g N U / n_D$	£41,835	£10,197	yes	year
H	Last-window MC ES95	£248,220	£75,001	yes	year
H'	Last-window MC VaR95	£224,246	£69,656	yes	year
I	GPD ES of daily ℓ_t (clamped)	£35,966	£7,102	no (V+H)	day
J	$p99(\ell_t) \times 252$	£1,005,366	£439,753	no	pseudo-year

Table 11. Master reconciliation. (E), (F) and (G) are the coherent family of one-year, margin-adjusted, full-sample functionals. (H) is the same family with an active-day mean and a last-window train slice. (I) is a daily POT mean above the 99th-percentile threshold. (J) is a stress exhibit.

13.4 Which number should a reader remember

If the question is “in a bad year, under the i.i.d. log-normal of Section 7, what is preventable dragon loss at the 95% expected-shortfall point, margin-adjusted, on the full sample?”, the answer is (F): about £60k gross view, about £12k netted view. If the question is “what is the expected margin-adjusted annual unfulfilled-dragon revenue, no tail, just a stretch of this year?”, the answer is (E): about £43k / £9k. If the question is “what does the last expanding window’s more aggressive MC say?”, the answer is (H): about £248k / £75k. None of these is a forecast of a real warehouse. They are three well-defined functionals of a simulated file, and they differ for reasons that are now algebraic rather than mysterious.

14. Data governance and WMS implications

14.1 The 80,995 / –80,995 pattern

StockCode 23843, PAPER CRAFT LITTLE BIRDIE, enters the gross file as $Q = 80,995$, $V = £168,469.60$, the sample maximum, and is followed on a short clock (on the order of twelve minutes in the raw file) by a negative-quantity row of the same magnitude. It is the canonical UCI outlier. Two interpretations are live.

Interpretation 1 — data-entry or integration error. A false large positive is booked and then reversed. The gross file then contains a phantom dragon. Any risk number that uses Scenario 1 without a matching rule will treat a keystroke as a £168k exposure. In a production WMS this is a data-quality incident: it should be caught by an order-entry validator (quantity caps, velocity checks, confirmation on outliers) long before it reaches a tail-risk engine. Scenario 2’s netting is a blunt annual analogue of that validator.

Interpretation 2 — genuine booked demand, subsequently cancelled. The warehouse may already have allocated inventory, reserved labour, or purchased inbound by the time the negative row arrives. Gross exposure is then the economically right object for the hours between book and cancel, and netting after the fact understates the operational tail. Working capital was committed against £168k, not against zero. A risk system that only ever sees netted tickets will never provision for that window.

The two-scenario design is the paper’s way of not choosing. A WMS with an audit trail would choose: it would timestamp allocation, pick, pack, ship and cancel, and the unfulfilled-dragon indicator of Section 6 would be an observed random variable rather than a simulated one. Until that file exists, reporting both the phantom-sensitive gross view and the netted view is the honest move.

14.2 A WMS KPI layer this project does not have

The standard execution KPIs that would close the loop between a dragon event and operations are order cycle time, perfect-order rate, fill rate, dock-to-stock time, inventory accuracy, pick accuracy, SLA attainment. Each of those is a time-stamped, system-of-record measurement. None of them is in UCI Online Retail. Attaching them to the dashboard is a product idea, not a result. The mathematical pipeline in this paper is the risk half of that product; the KPI half requires proprietary telemetry the author does not have.

14.3 What would change in a live WMS

Dragon would be an observed predicate (value and cycle-time jointly exceeding a policy threshold), not a weighted sample. Delay D_i would be pick-complete minus order-entry, not a clipped log-normal. The surge flag would be an observed capacity state (wave, labour, carrier), and the ATE of surge on cycle time would become a genuine causal question, identified by whatever operational instrument is available. QTE of dragon on value would still be of interest, but it would be a description of a policy-defined set rather than a restatement of the sampling weights. Hawkes would be fitted to observed event times with a kernel whose time unit matched the clustering scale of interest. Backtests would run daily, on rolling proprietary history, with enough exceptions for Kupiec to have power.

15. Limitations, and what a production version would change

This section records the limits of the work, not an apology. Every item was accepted at the outset as a consequence of public data, undergraduate compute, and a one-year retail file.

15.1 Structural

Delays are synthetic. No result about SLA, unfulfilment, holding, Monte Carlo, Hawkes timestamps, or backtest realised loss is an observation of a warehouse. Dragon labels are assigned as a function of value; QTEs are partly mechanical; ATEs of surge are a negative control. Data vintage is 2010–2011: fulfilment technology, carrier networks and customer expectations have moved. GPD and Hawkes are fitted on the full sample, not purged. EVT standard errors ignore the daily dependence of Section 4.3. One year of tickets gives 34 backtest windows; coverage tests have low power. Five-year arithmetic ($5B$, $5 \times ES$) assumes a constant dragon rate, margin and SLA policy.

15.2 Modelling choices, as implemented

Monte Carlo paths are i.i.d. log-normal; Hawkes clustering is not injected (no Ogata thinning of the annual paths). CV is capped at 3; the MC module uses a calendar-day mean with an active-day dispersion, while the backtest uses an active-day mean — two different functionals, reconciled in Section 13. PSM is a 50,000-row penalised logit with caliper 0.005 and no love plot / double robustness; Scenario 1 matching may reuse controls, Scenario 2 does not; QR is a single 180,000-row subsample. Kalman Q , R are fixed; the monthly forecast is clipped to $\{6, \dots, 10\}$. Hawkes follows the undecayed-jump recursion of Section 9.2. The daily-loss GPD is fitted above the 99th percentile of realised loss, and the printed 95% VaR/ES pair uses $\alpha = 0.05$ with $VaR = \max(u, \cdot)$, so it is a far-tail daily mean excess. Quantity never receives a GPD MLE; AMSE Hill is a threshold diagnostic. Serial dependence is left in the series: there is no declustering.

15.3 What I would do next, given a WMS extract

Integrate Hawkes intensity into the Monte Carlo by thinning: draw each day's dragon count from a time-changed Poisson rather than from an i.i.d. log-normal revenue. Re-fit GPD and Hawkes inside the purged window, not on the whole sample. Estimate Q , R by EM or by a proper Bayesian structural time

series if a posterior is actually wanted. Define dragon from observed cycle time $\times$ value, so that QTE is a contrast of a policy set rather than of a weighted sample. Replace the four mitigation multipliers by a small operations model (safety stock on the SKUs that actually dominate U). None of that is in this repository, and claiming it is would be the wrong kind of completeness.

16. Conclusion

The warehouse tail-risk problem, as posed here, is a problem in extreme value theory with a risk-management translation layer. Quantity on high-volume SKUs is in the Fréchet domain; three tail-index estimators and a positive mean-excess slope agree on that. Because the public file has no fulfilment clock, delays and dragon labels are assigned, and the only honest causal statement is: the surge ATE on net revenue is a negative control (empirically near zero), and the QTE of the dragon grouping on order value is the descriptive object of interest (£3.7k at the 95th, £156k at the 99.9th, gross view). Annual preventable loss is then a Monte Carlo functional of dragon revenue, an empirical breach rate, and a 30% margin; its full-sample $ES_{0.95}$ is about £60k (gross) and £12k (netted). A Kalman local-linear-trend filter projects a clipped monthly count; a Hawkes MLE on minute-scale timestamps is maximised under the undecayed-jump recursion of Section 9. A purged expanding-window backtest, with the coverage tests written out, is the right validation idea applied to a sample too short to validate much.

The paper's usable contribution is not a sterling number to take to a CFO. It is a chain of identities—from regularly varying tails, through an assigned delay, to a family of loss functionals that have been written down and reconciled.

What this monograph demonstrates, and what it does not.

It demonstrates that the full mathematical stack (EVT on quantity, POT/GPD on daily realised loss, log-normal MC, QTE, Hawkes MLE, Kalman filtering, Kupiec/Christoffersen) can be derived and wired, end to end, on a public retail file, with every sterling headline identified as a named functional. It does not demonstrate a validated operational model, a real SLA effect, or a policy prescription to hold 99th-percentile service on dragons. Those claims would require observed delays.

Appendix A. Notation, configuration, and additional derivations

A.1 Notation

Symbol	Meaning
V_i, Q_i, P_i	order value, quantity, unit price of ticket i
T_i, D_i	assigned delay tier; delay in minutes
δ_i, W_i, U_i	dragon flag; surge flag; unfulfilled-dragon flag
H_i, R_i^{net}	holding cost; net revenue $V - H$
ξ, u, β	EVT index; threshold; GPD scale
μ, α, β, n	Hawkes baseline, jump, decay, branching ratio α/β
μ_ℓ, σ_ℓ	log-normal location and scale of delay
μ_d, σ_d	log-normal location and scale of daily dragon revenue
h, g	annual holding rate 0.25; gross margin 0.30
L, ℓ_t	annual MC loss (margin-adjusted); daily realised loss ($V+H$)
$\Delta(q), \pi$	QTE of dragon on V at quantile q ; dragon mean premium
I, B	causal annual impact $g(365/\Delta)U$; Kalman bleed $g N U / n_D$

Table 12. Notation used in the body. Subscripts ℓ, d distinguish delay-log-normal and daily-revenue-log-normal parameters; Hawkes (μ, α, β) and GPD (ξ, β) share letters by convention and are always named in context.

A.2 Scenario configuration, as in config.py

Parameter	Scenario 1	Scenario 2
SURGE_PCT	0.18	0.18
DRAGON_PCT	0.00032	0.00025
DRAGON_BIAS_EXP	1.35	1.20
NORMAL / SURGE / DRAGON mean (min)	25 / 90 / 420	25 / 90 / 360
$\sigma_\ell N / S / D$	0.50 / 0.65 / 0.85	0.50 / 0.65 / 0.82
Clip $N / S / D$ (min)	[5,120] / [30,480] / [180,1440]	[5,120] / [30,360] / [200,1440]
SLA bins (min)	[0, 30, 120, 480, ∞)	[0, 30, 120, 360, ∞)
SLA_BREACH_MIN	240 min (4 h)	360 min (6 h)
ANNUAL_HOLDING_RATE h	0.25	0.25
GROSS_MARGIN g	0.30	0.30
TARGET_ANNUAL_DRAGONS	104	72
MIN_TRAIN_DAYS / STEP / PURGE	90 / 7 / 3	90 / 7 / 3
PSM caliper / PS subsample	0.005 / 50,000	0.005 / 50,000
QR subsample	180,000	180,000
QUANTILES	[0.95, 0.999]	[0.95, 0.999]
RNG seed	314159	314159

Table 13. All of these are assumptions, not estimates from warehouse telemetry. WERC/CSCMP public summaries inform the delay means, holding rate and SLA windows. Dragon rates (3.2 bps / 2.5 bps) and bias exponents put a thin, expensive tail in the dragon

bucket; they are not a 99th-percentile cutoff on value.

A.3 Score equations of the GPD

For completeness, the GPD scores used by any gradient-based MLE (the backtest uses L-BFGS-B on $-\ell$) are, writing $z_j = Y_j / \beta$ and $\phi_j = 1 + \xi z_j$,

$$\frac{\partial \ell}{\partial \xi} = \frac{1}{\xi^2} \sum_j \log \phi_j - (1 + 1/\xi) \sum_j \frac{z_j}{\phi_j}, \quad (68)$$

$$\frac{\partial \ell}{\partial \beta} = -\frac{N_u}{\beta} + (1 + 1/\xi) \sum_j \frac{\xi Y_j / \beta^2}{\phi_j}. \quad (69)$$

At $\xi = 0$ the exponential scores take over by continuity. The code rejects $\xi \geq 1$ (barrier plus box bound) because $E[Y] = \beta/(1-\xi)$ is then infinite, without which ES is undefined. That identity is also why the mean-excess function of an exact GPD is $(\beta + \xi u)/(1-\xi)$.

A.4 Hill variance and a caution on $\sqrt{k}$

In the exact Pareto model, $\text{Var}(\xi_{\text{Hill}}) \approx \xi^2/k$. With $k = \lfloor \sqrt{n} \rfloor \approx 630$ and $\xi \approx 0.5$ this is a standard error of about 0.02, which would make 0.515 look precise. The model is not exact Pareto, the sample is dependent, and $k = \sqrt{n}$ is a rule of thumb (not Hall's AMSE-optimal k). The paper therefore never quotes a confidence interval around Hill on this file. The AMSE curve of Section 5.11 is a stability diagnostic of a different Hill-type statistic (excess-Hill, $k = N_u$), not a standard error for (17), and u^* is not used to fit a GPD on quantity.

A.5 Why last-window ES exceeds full-sample ES

Let f be the fraction of calendar days that are dragon-active. The MC module uses $E[R] = f \cdot \mu_{\text{active}}$; the backtest uses $E[R] = \mu_{\text{active}}$. The annual sum then scales by $1/f$. With 127 dragons in 374 days, if each dragon-day carried one dragon, $f \approx 0.34$ and $1/f \approx 3$. A ratio of $248,220 / 60,134 \approx 4.1$ is in that neighbourhood, with the residual explained by a different training slice, a different r_v , and the CV cap. The ratio is a mean-definition effect, not a discovery that the last window was four times riskier.

A.6 Seed

Every numpy generator in the delay, Monte Carlo, backtest, Shapiro–Wilk and AMSE stages is constructed as `default_rng(314159)`. The recorded run is therefore a single point in the seed space for the loss functionals. A production stress test would report a distribution over seeds; this monograph reports the number the repository actually emitted.

A.7 Additional empirical exhibits

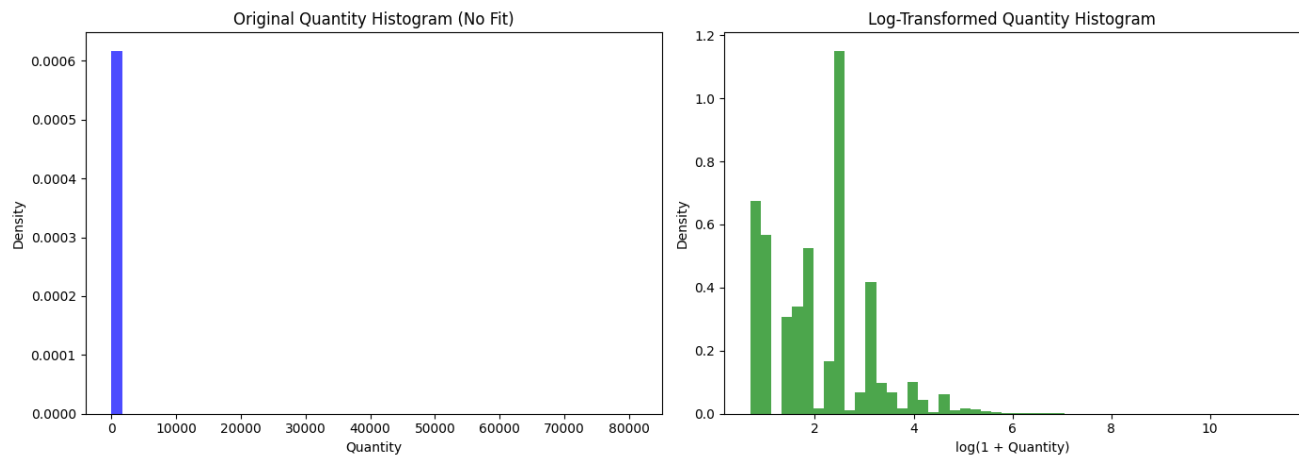


Figure 15. Quantity on the original and $\log(1+Q)$ scales, Scenario 1, Pareto-filtered sample. The left panel is unusable as a density; the right panel is still right-skewed.

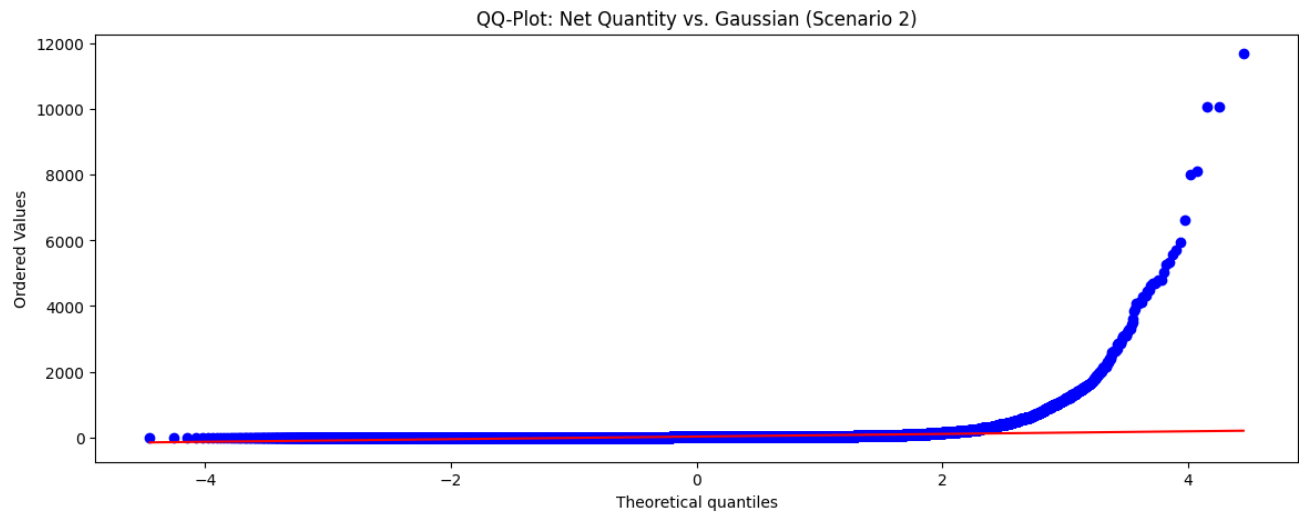


Figure 16. Normal QQ-plot of net quantity, Scenario 2. The right-tail bend survives netting.

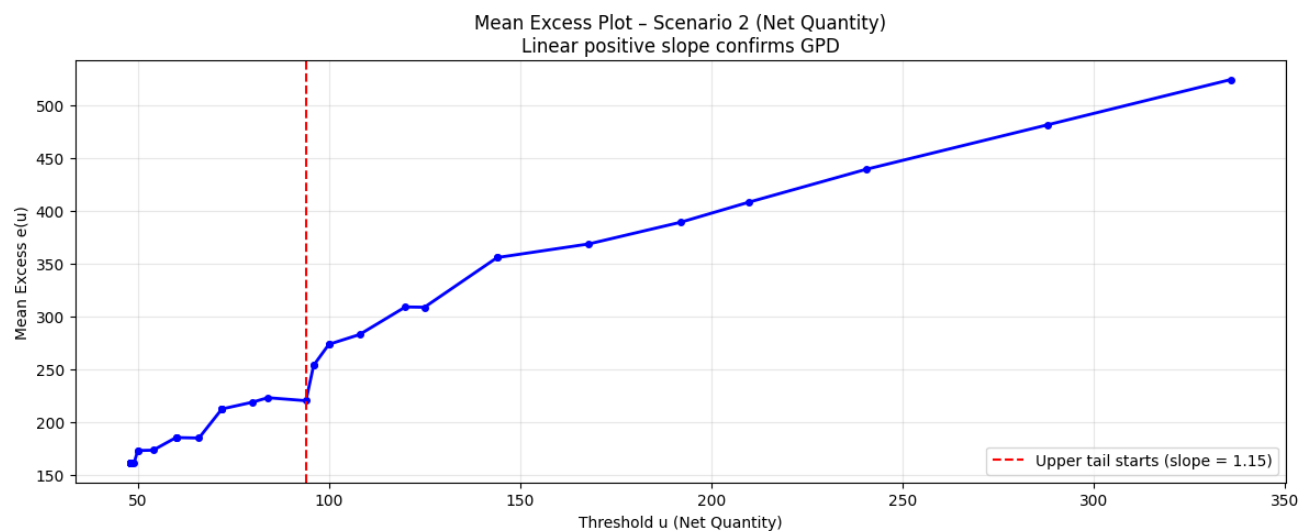


Figure 17. Mean-excess plot, Scenario 2. The upper-tail slope is positive (+1.089 on the recorded run): the Fréchet signature survives netting.

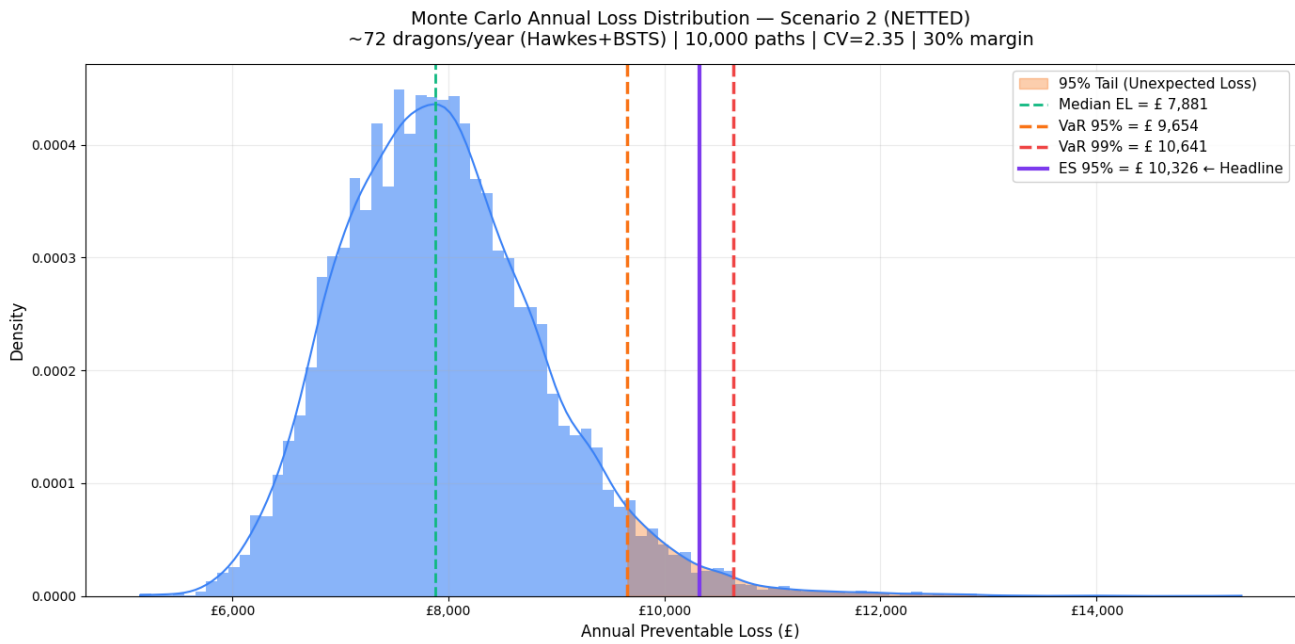


Figure 18. Monte Carlo annual loss density, Scenario 2. Same construction as Figure 8, smaller scale after netting. Table 5 is the recorded-run table.



Figure 19. Four-panel risk report, Scenario 2 (2/33 violations). This is the more informative coverage test of the two scenarios.

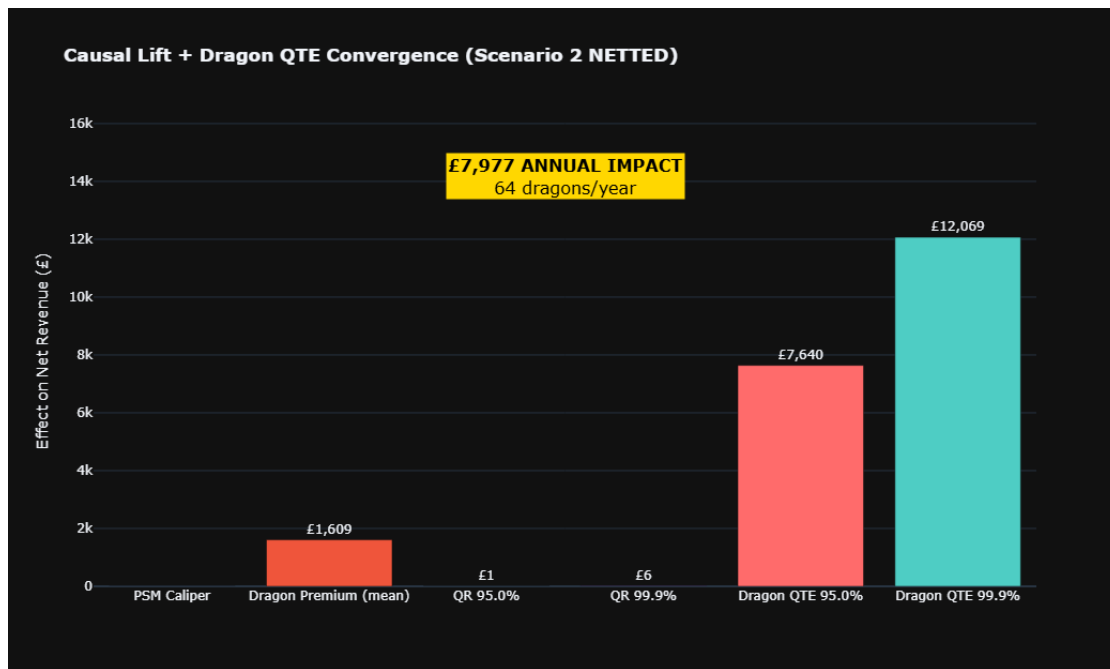


Figure 20. Scenario 2 causal panel. Left-of-zero surge ATE is the negative control; dragon premium and QTEs after netting are the object of interest.

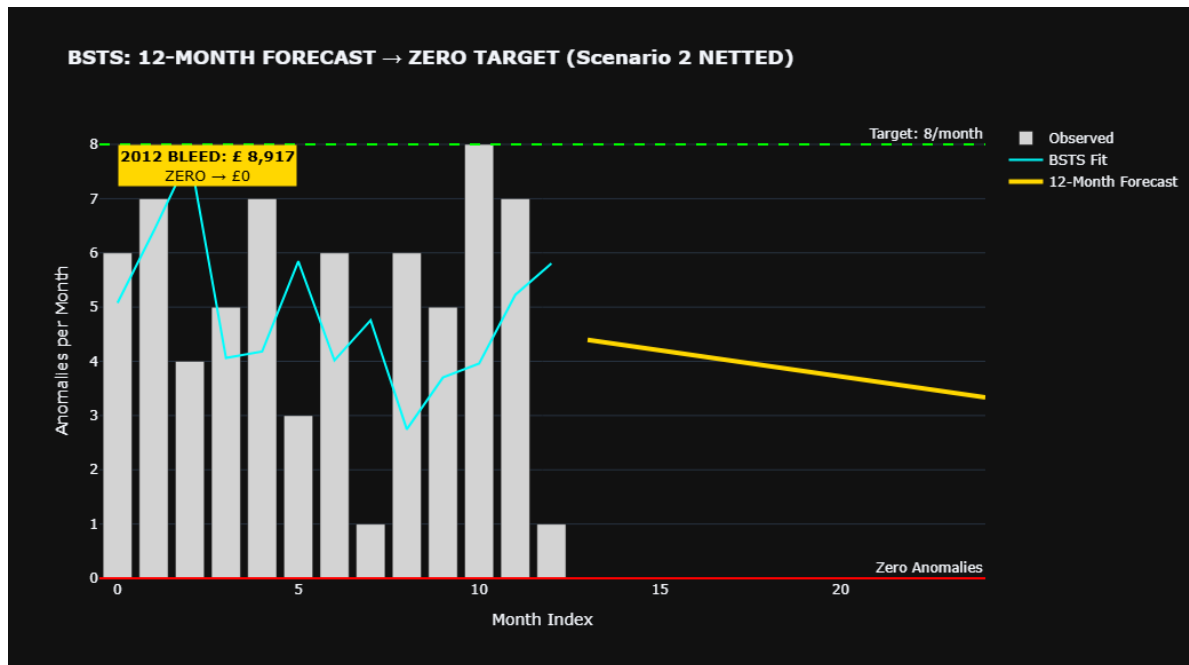


Figure 21. Kalman local-linear-trend fit and 12-month projection, Scenario 2. The printed 2012 count is the floor of the $\{6, \dots, 10\}$ clip, times 12.

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Licence: Apache 2.0 on the companion code. This monograph is independent research. It is not operational advice. Last compiled August 2026.

A note on implementation variants

Where an estimator admits more than one standard implementation, the version used here is the one in the companion code. Those choices are small, documented departures from a single textbook presentation; they are not a claim that the textbook form is incorrect.

Readers who identify a clear mathematical or logical error are invited to write to shridathjoshi@gmail.com.

— Pranav